



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Trends in Computer Vision Research and Applications

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Key Process Technology in Manufacturing



**Cognex DataMan
(1982)**

**Inspection, Gauging,
Guidance, *Tracing*,
*Compliance***

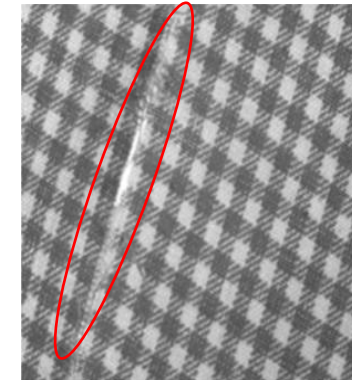


Whole Process Optimization

Automotive



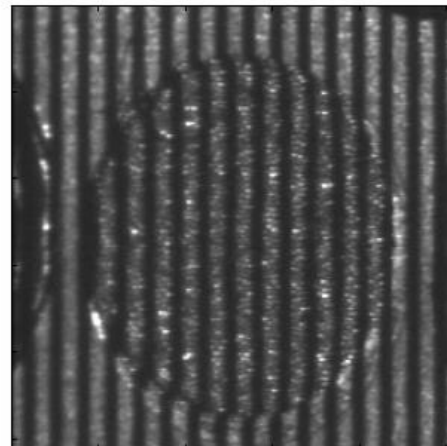
Textile



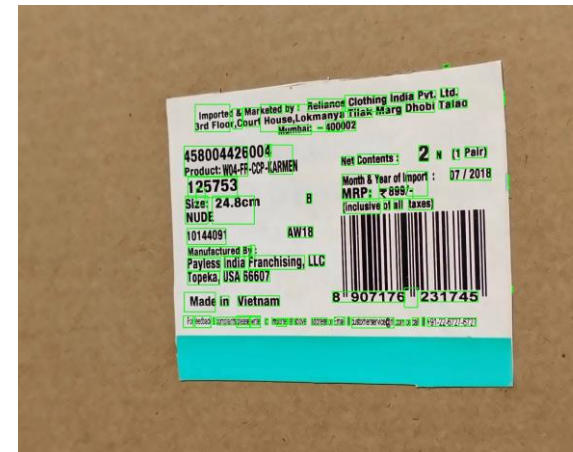
Food&Bevarage



Electronics

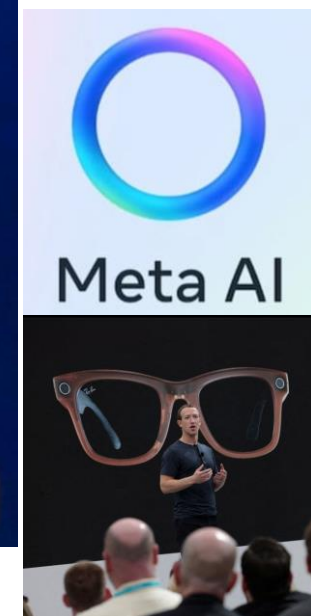
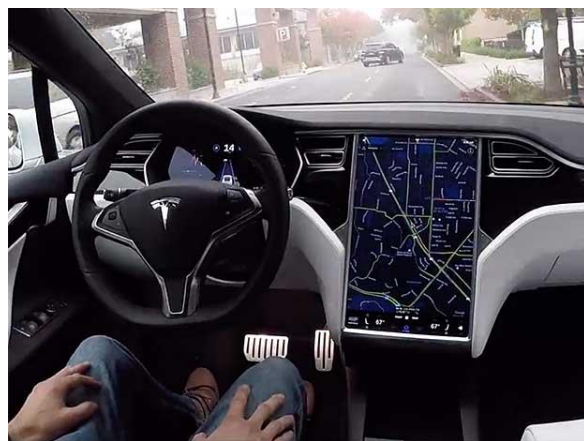


Transportation & Logistics

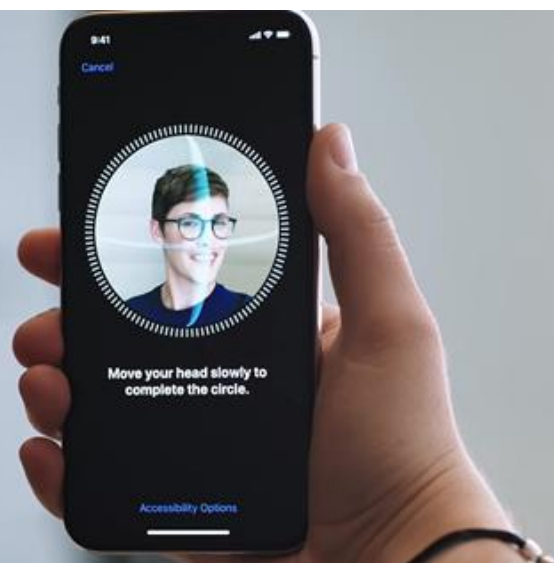
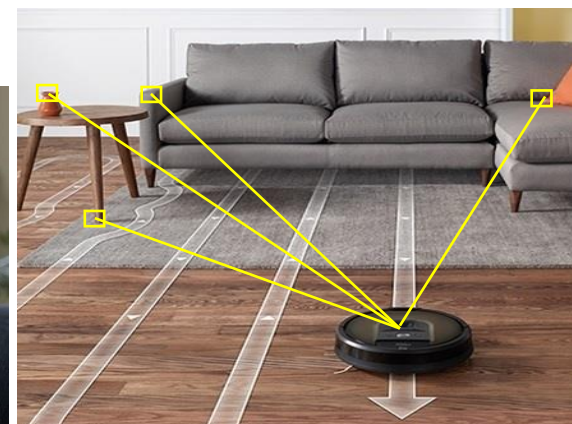


Pharma & Tobacco

Mass-Market Consumer Products



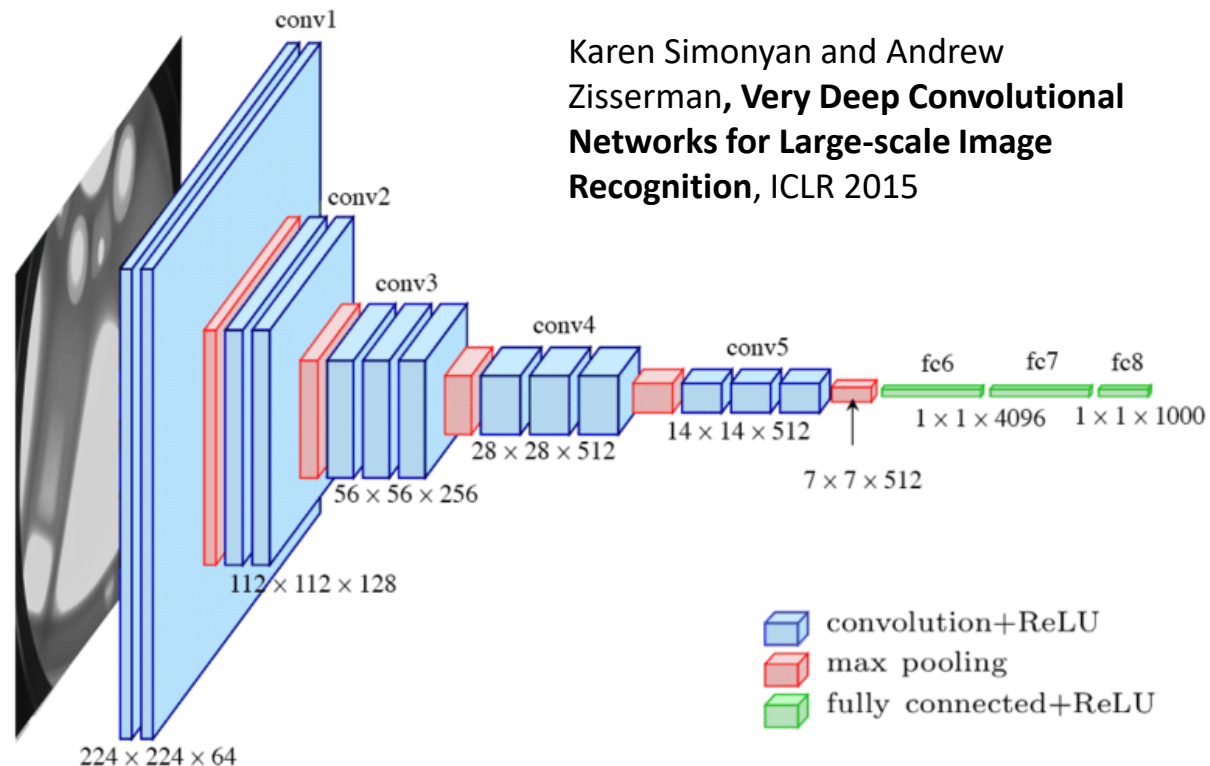
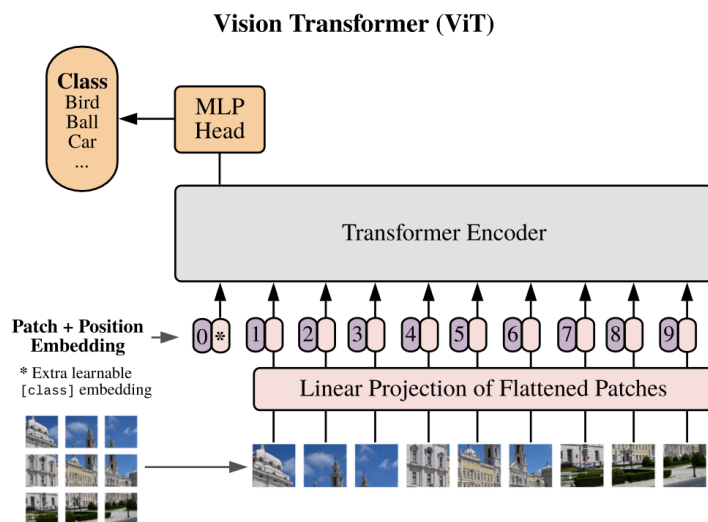
Google



Deep Learning (aka AI)

Deep learning is representation learning !

A Krizhevsky, I Sutskever, and GE Hinton,
Imagenet classification with deep convolutional neural networks. NIPS 2012.



Karen Simonyan and Andrew Zisserman, **Very Deep Convolutional Networks for Large-scale Image Recognition**, ICLR 2015

Alexey Dosovitskiy et. al., **An image is worth 16x16 words: Transformers for image recognition at scale**. ICLR 2020.

2012

2020

Impact of computer vision research



	Publication	<u>h5-index</u>	<u>h5-median</u>
1.	Nature	<u>488</u>	745
2.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	<u>440</u>	689
3.	The New England Journal of Medicine	<u>434</u>	897
4.	Science	<u>409</u>	633
5.	Nature Communications	<u>375</u>	492
6.	The Lancet	<u>368</u>	678
7.	Neural Information Processing Systems	<u>337</u>	614
8.	Advanced Materials	<u>327</u>	420
9.	Cell	<u>320</u>	482
10.	International Conference on Learning Representations	<u>304</u>	584



Novel-View Synthesis (Radiance Field)



<https://lumalabs.ai/>



NeRF, 2020



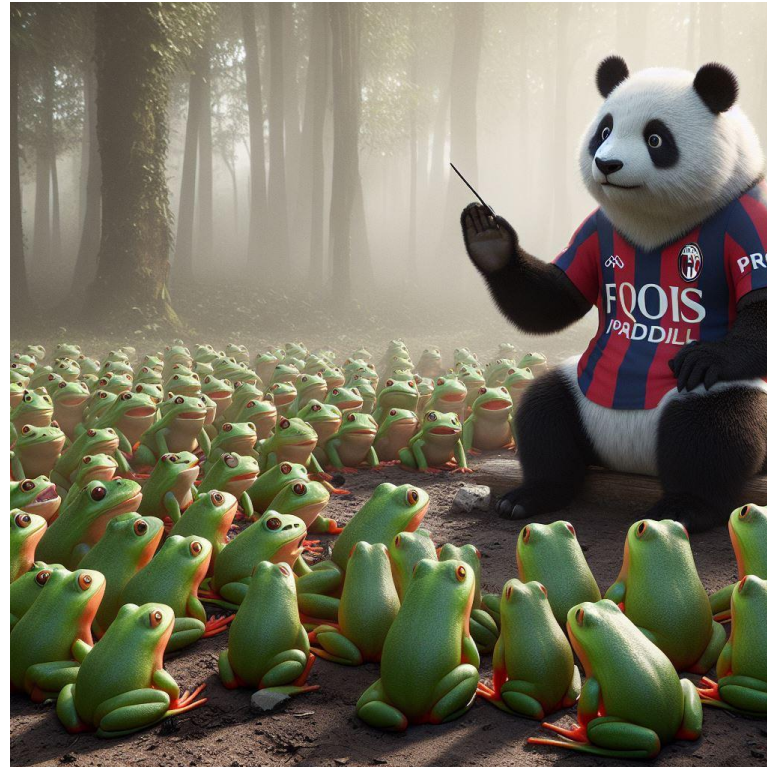
3D Gaussian Splatting, 2023

More Generative Tasks

DDPM aka Diffusion Models, 2020



Latent Diffusion Models,
2022 (Stable Diffusion)



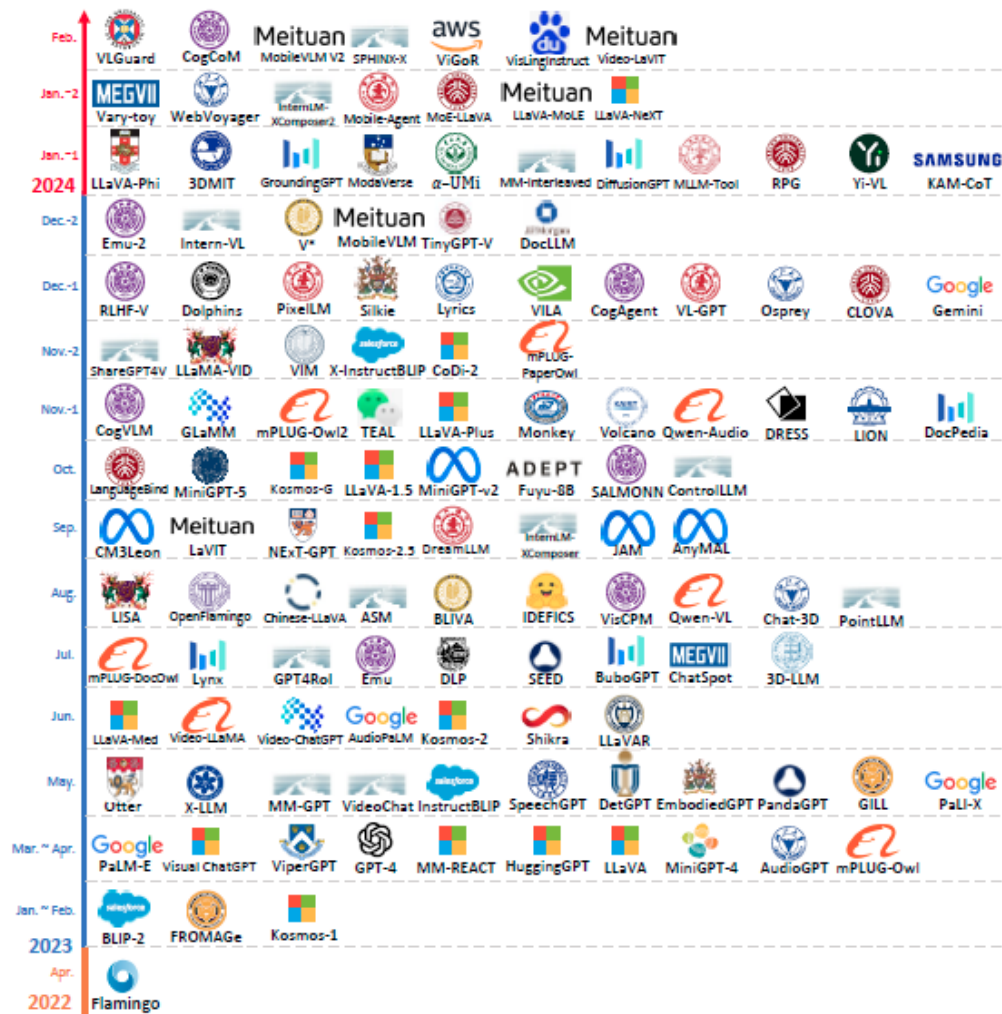
A photorealistic image of a panda wearing
the jersey of Bologna FC and teaching to a
group of very interested frogs

DALL-E 3 (Bing)

Genie (Luma AI)

A panda wearing the jersey
of Bologna FC

MLLMs (Multimodal Large Language Models)



Timeline of MLLMs

(<https://arxiv.org/pdf/2401.13601>)

Visual input example, Extreme Ironing:



LLaVA: Large Language and Vision Assistant

Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User
LLaVA

What is unusual about this image?

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

[Start a new conversation, and clear the history]

User
LLaVA

What's happening in the scene?

The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention **due to his unconventional choice of ironing his clothes on top of a moving car**. The city street around him is bustling with activity, adding to the unique nature of the scene.

User
GPT-4 [32]

What is unusual about this image?

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

User
BLIP-2

Can you explain this meme in detail?

a man is sitting on the back of a yellow cab

User
OpenFlamingo

Can you explain this meme in detail?

The man is drying his clothes on the hood of his car.

Deep Learning (aka AI) ? It's happening



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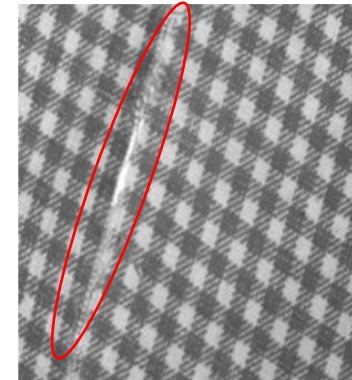


Whole Process Optimization

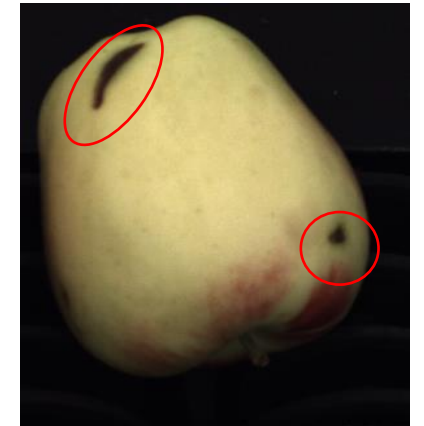
Automotive



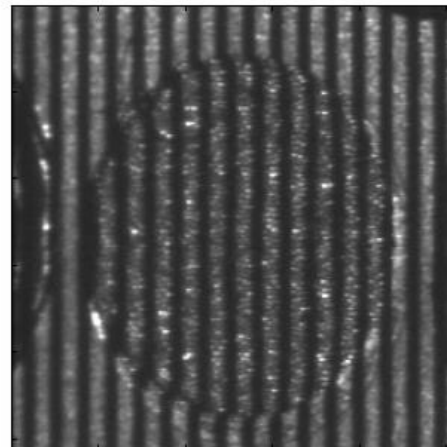
Textile



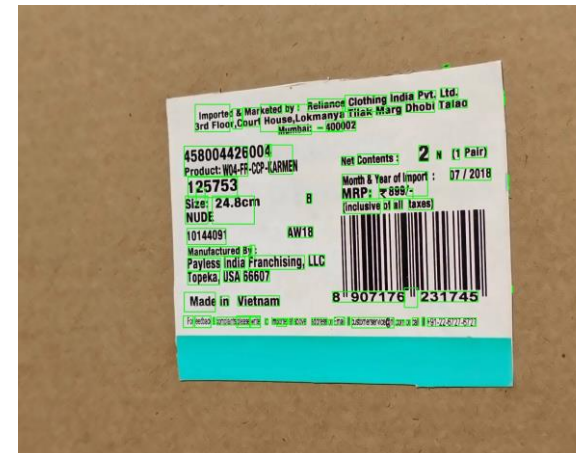
Food&Bevarage



Electronics



Transportation & Logistics



Pharma & Tobacco

Industrial Anomaly Detection (IAD)

Good (nominal) samples are abundant and easy to collect, whereas defects are rare and (often) unpredictable

➔ Learn to model good samples only !



Nominal Samples



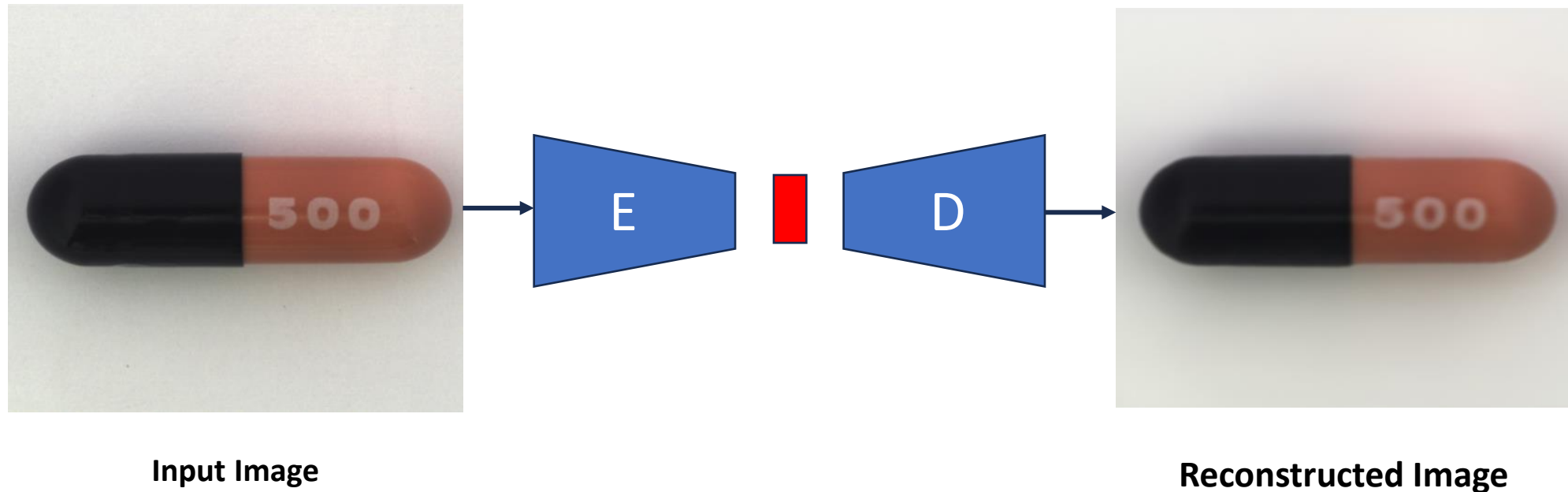
Anomalous Samples



Anomaly Segmentation

MVTec AD
Dataset [1]

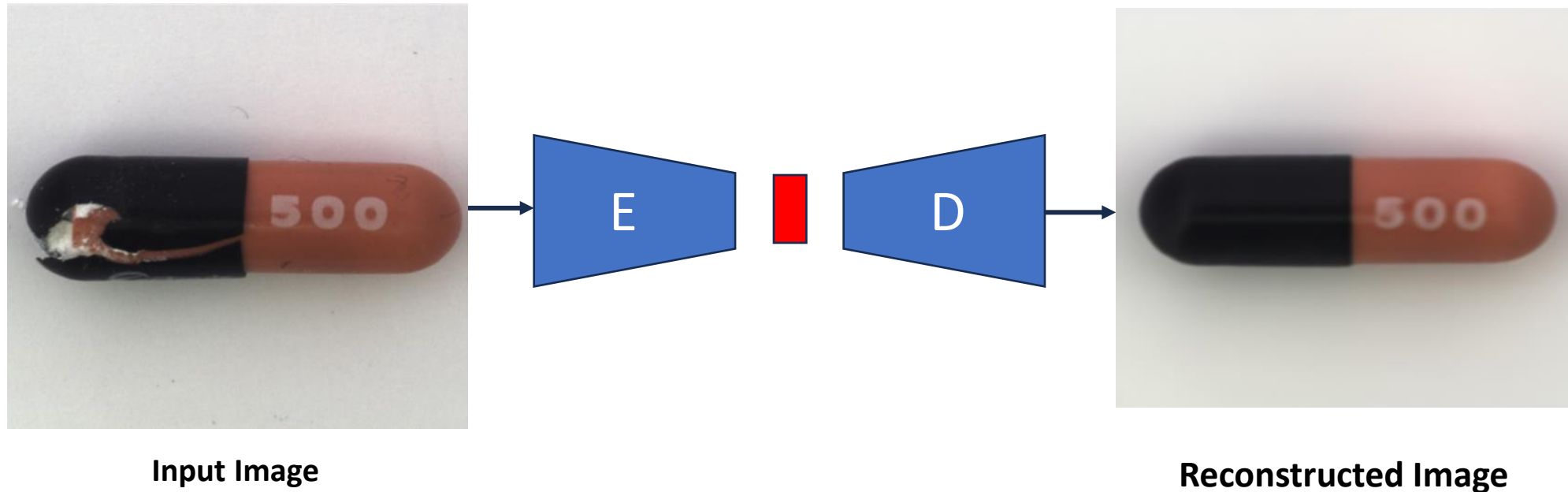
Learning to reconstruct good samples (1)



AutoEncoder Network

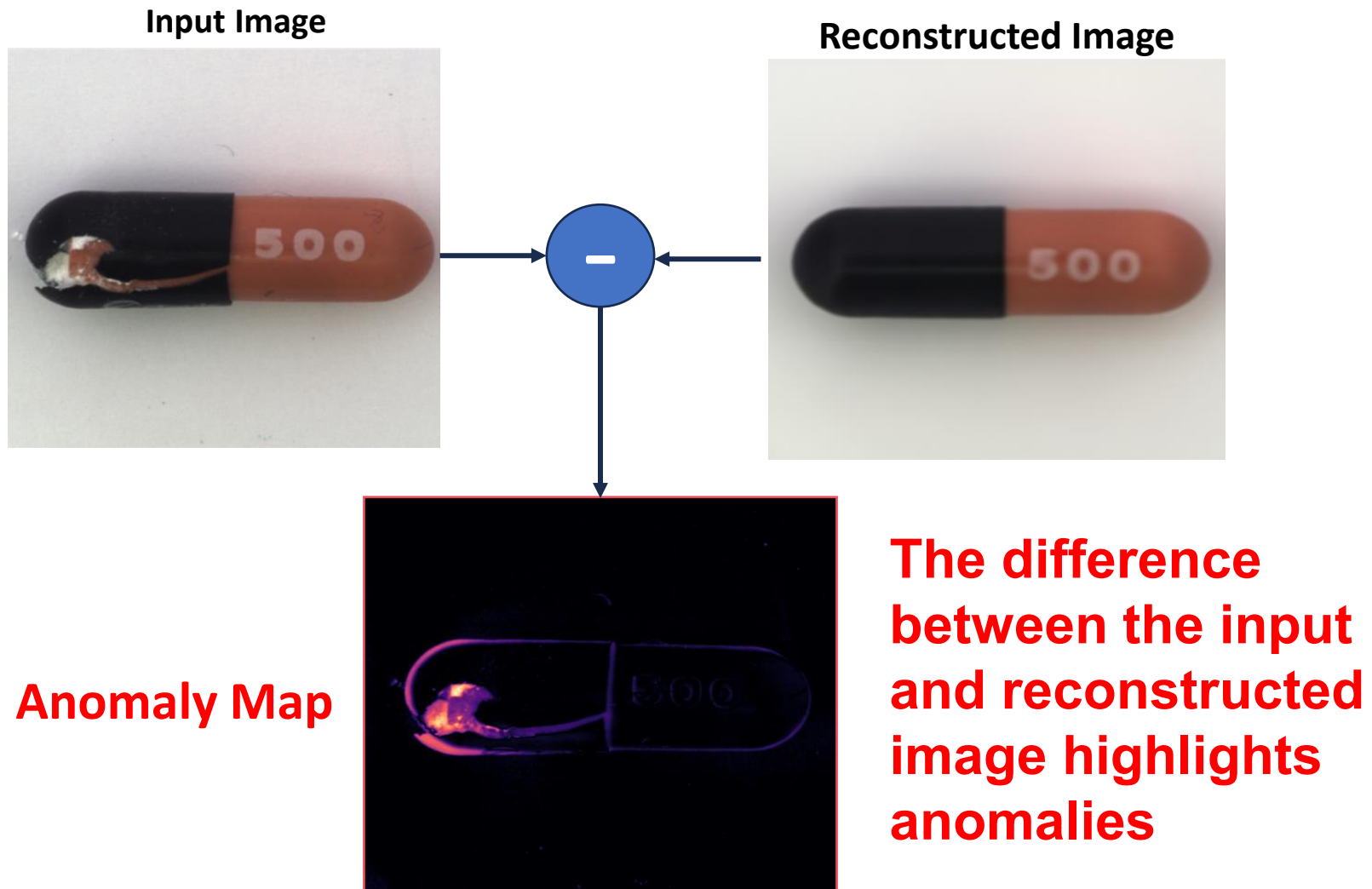
Encoder-Decoder architecture trained to *reconstruct* the input images

Learning to reconstruct good samples (2)



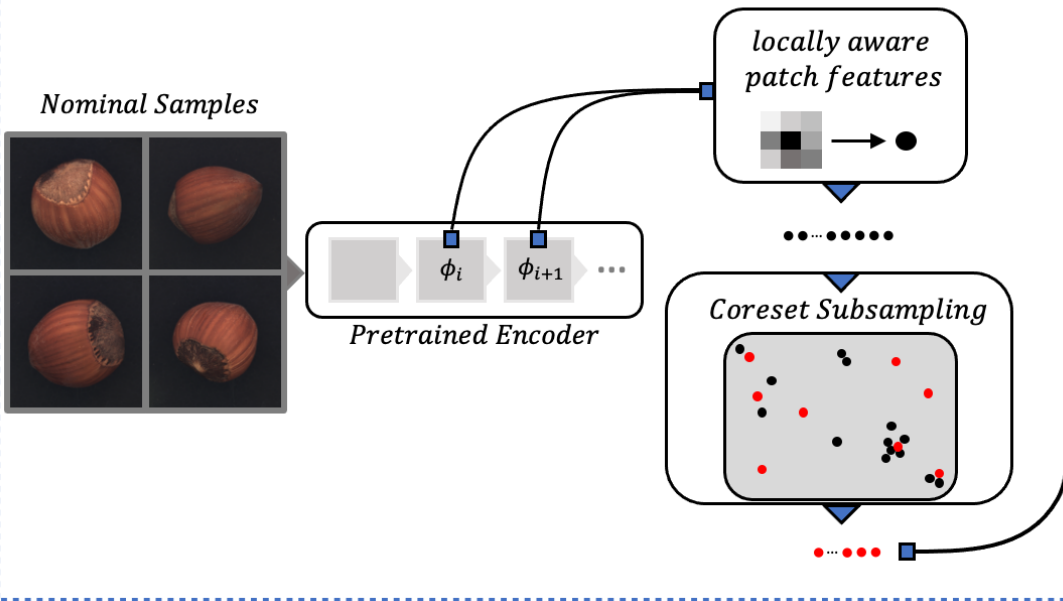
The model learns to reconstruct good samples only. Hence, when provided with an anomalous sample, the reconstructed image looks different from the input one.

Learning to reconstruct good samples (3)

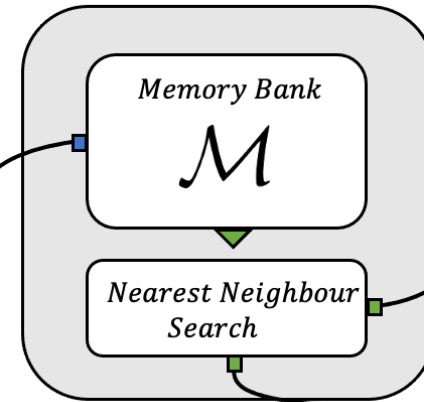


State-of-the-art Anomaly Detection: PatchCore [2]

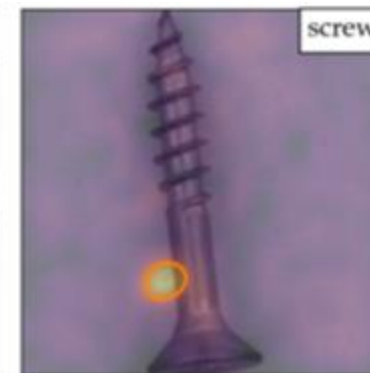
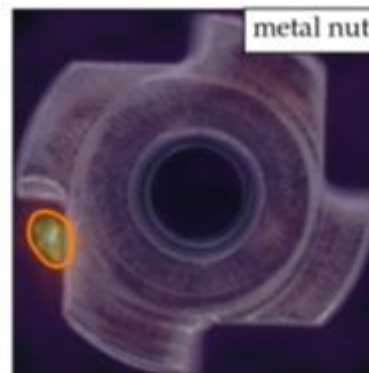
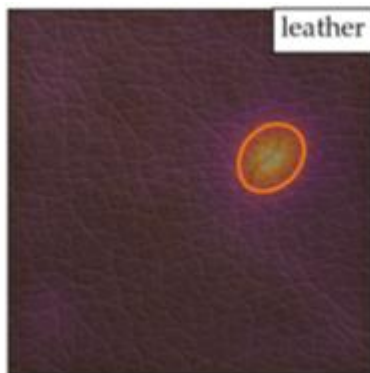
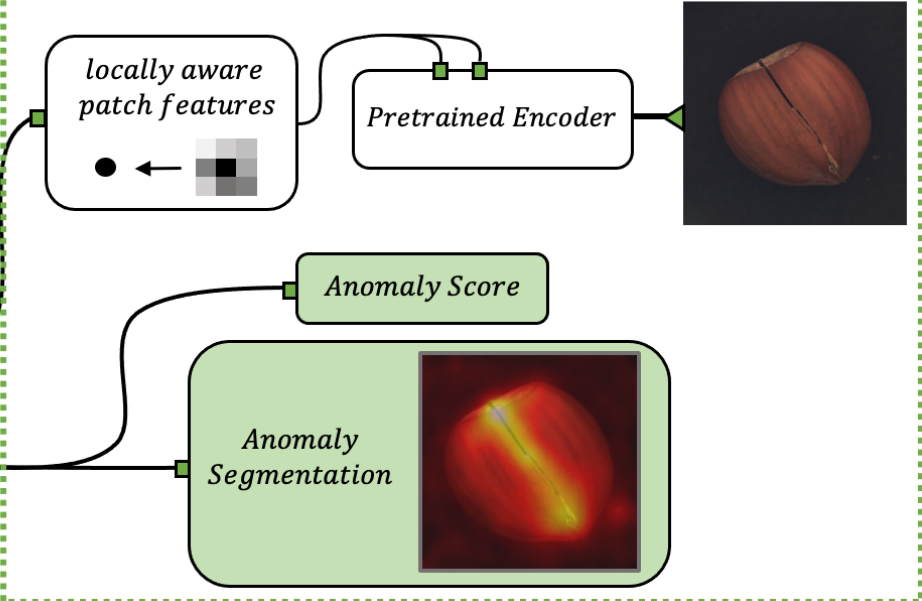
Training



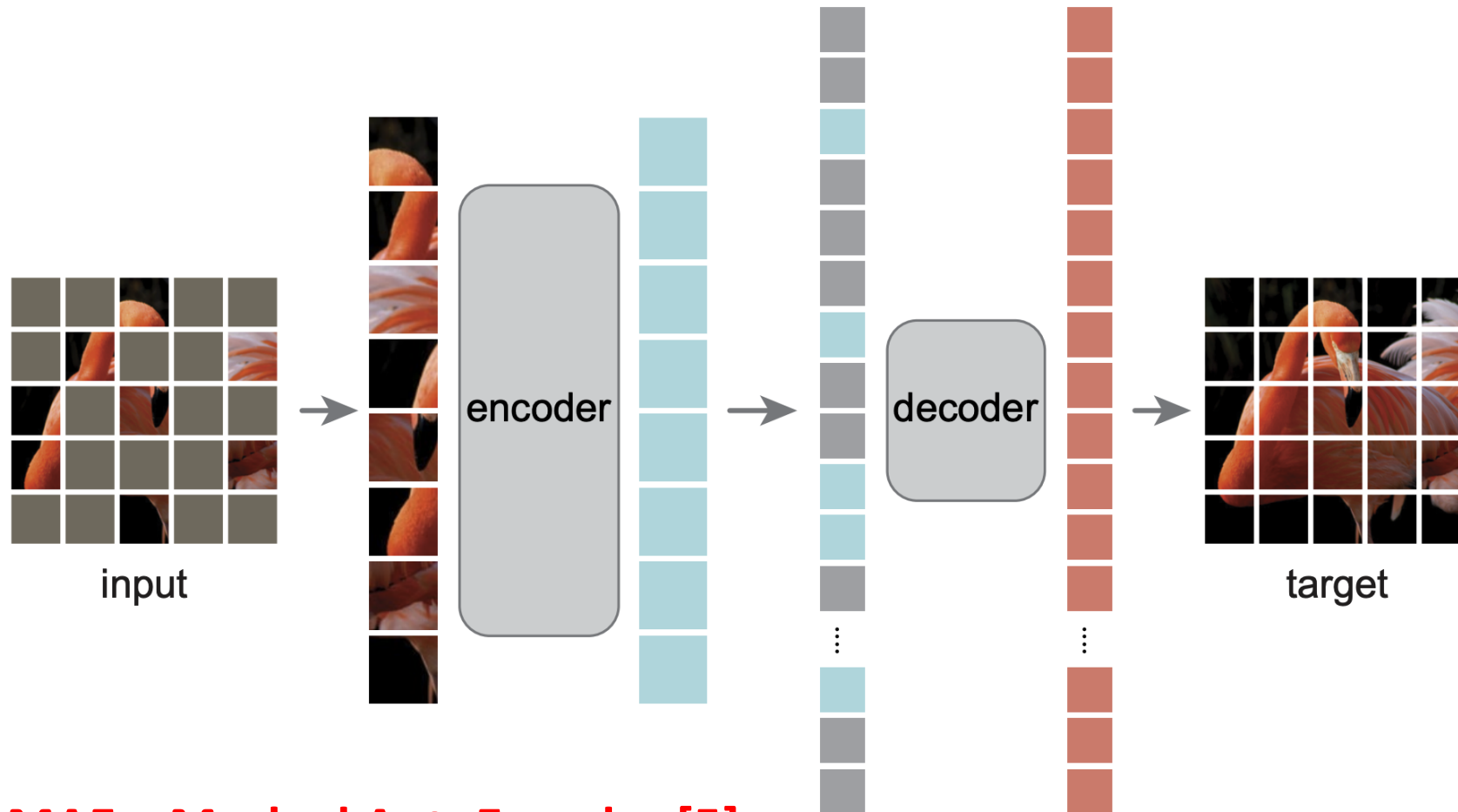
PatchCore



Testing



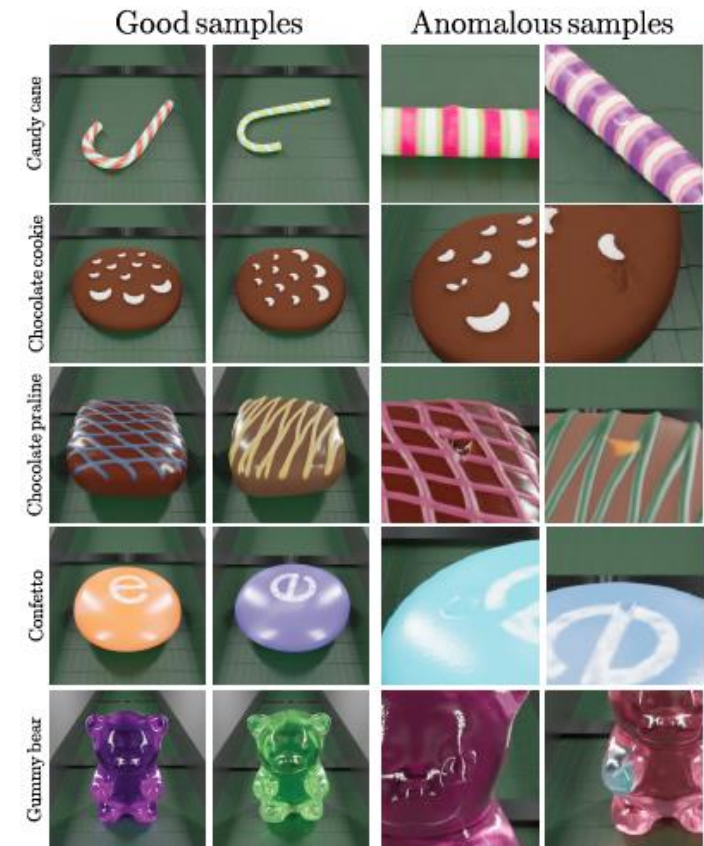
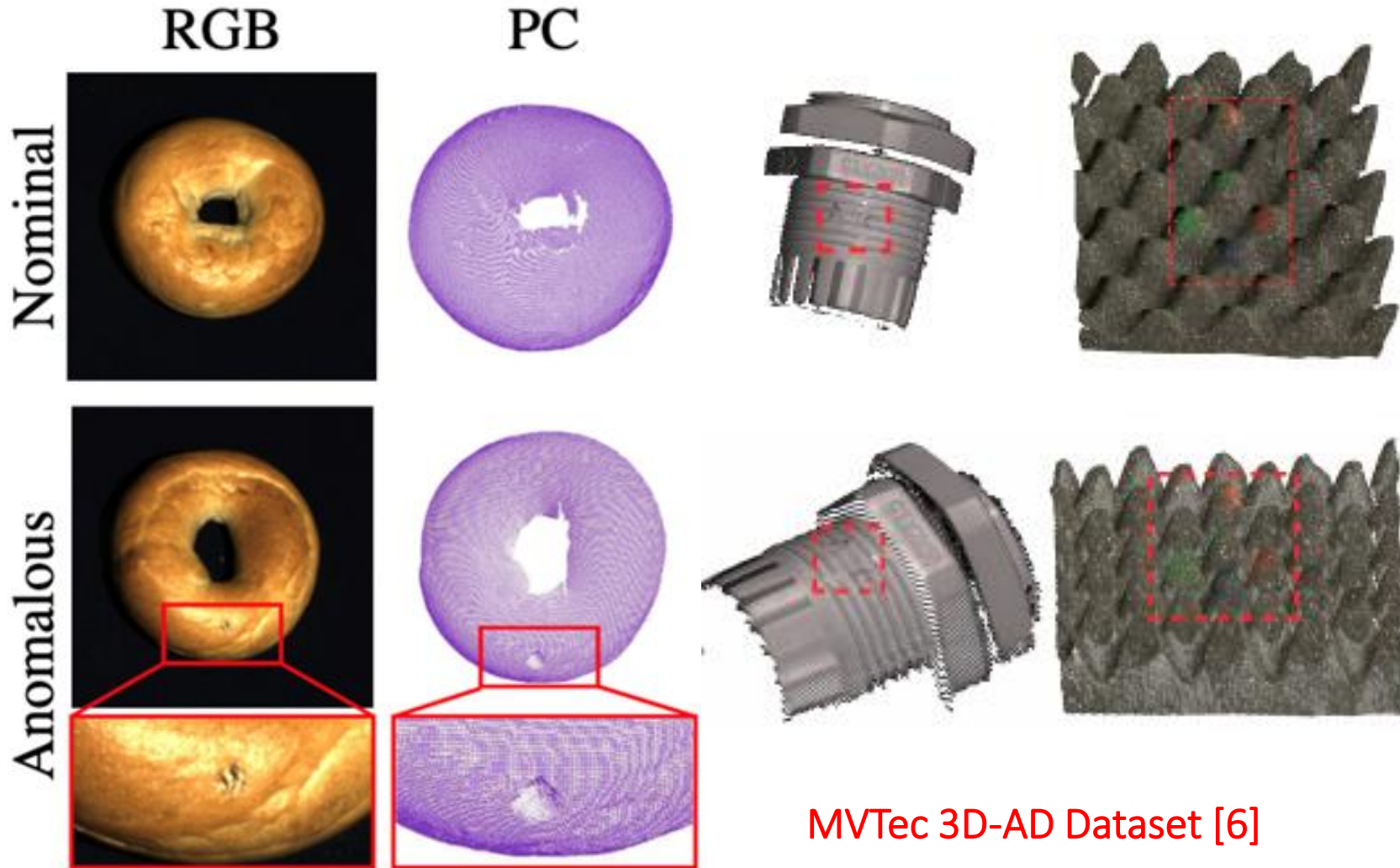
Self-Supervised Learning and Vision Foundation Models



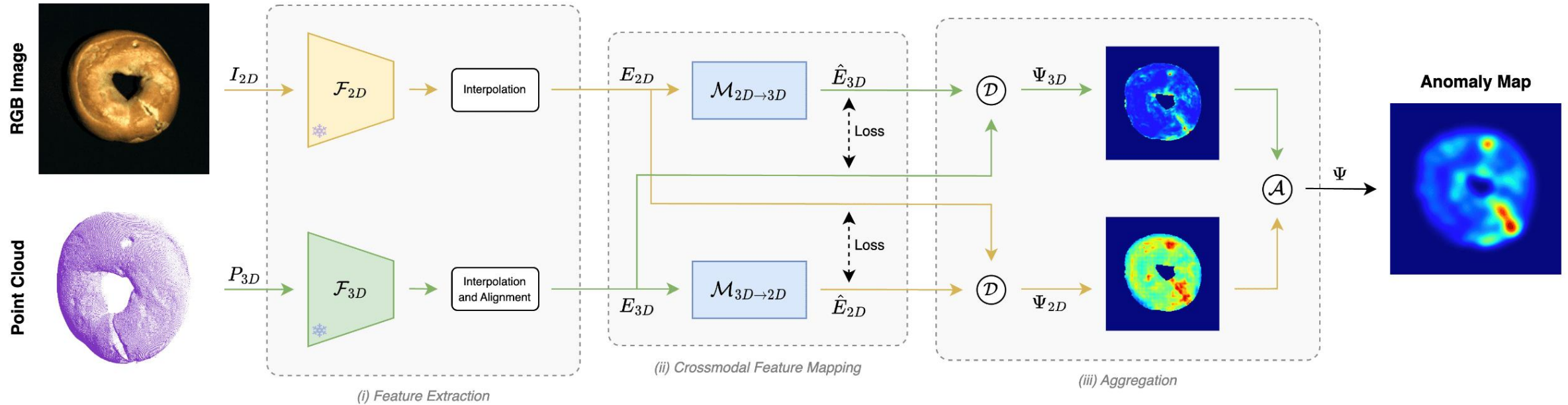
MAE – Masked AutoEncoder [5]

- *Vision Foundation Models (VFM) are image encoders trained on huge unlabelled datasets by **self-supervised learning**.*
- VFM provide high-quality and general-purpose features to be used without any further training for a variety of diverse downstream tasks.
- Prominent VFM are DINO [3,4] and MAE (left).

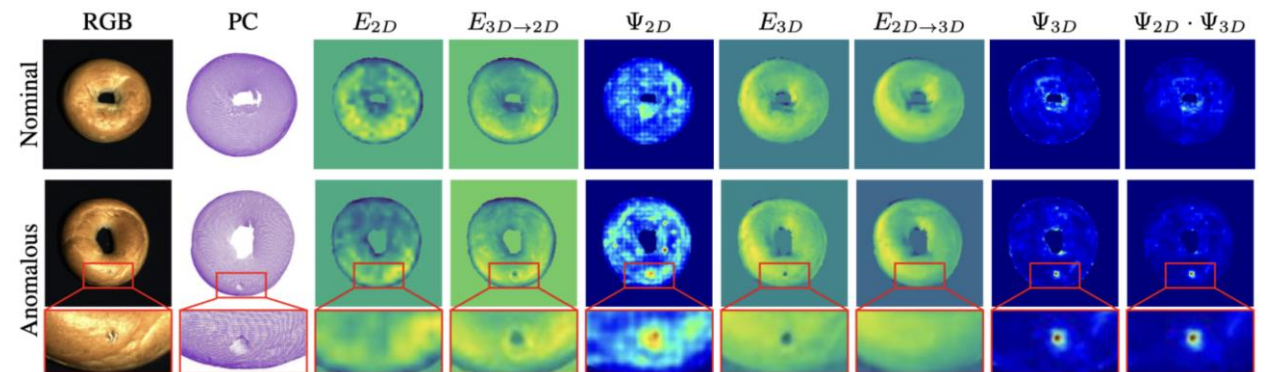
Industrial Anomaly Detection goes Multimodal (RGB+3D)



Crossmodal Feature Mapping [8]



- Two lightweight MLPs are trained solely with nominal samples so to predict foundational features across the two modalities.
- At inference time, the differences between predicted and actual features are aggregated into the final anomaly map.

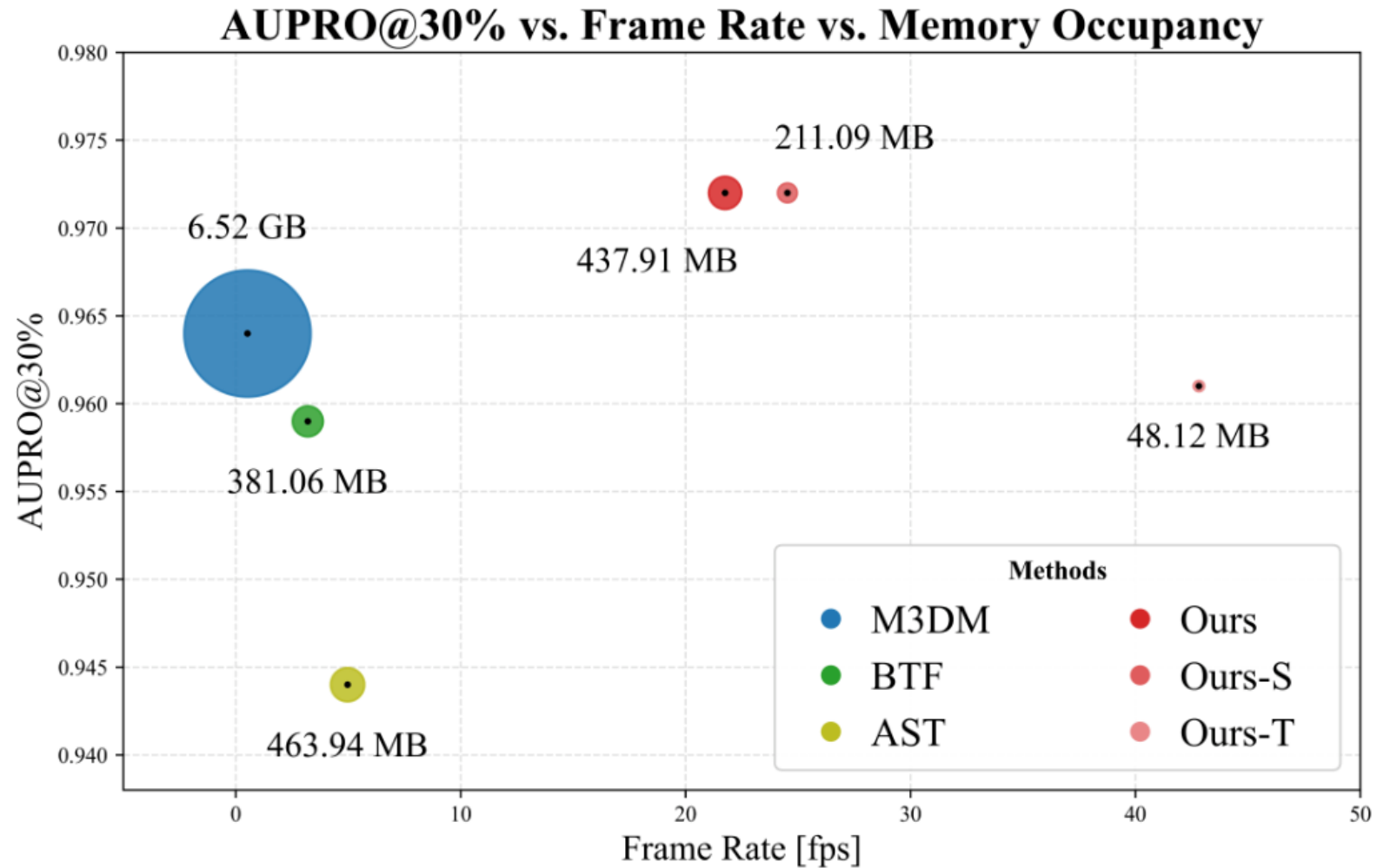


Quantitative results on MVTec 3D-AD

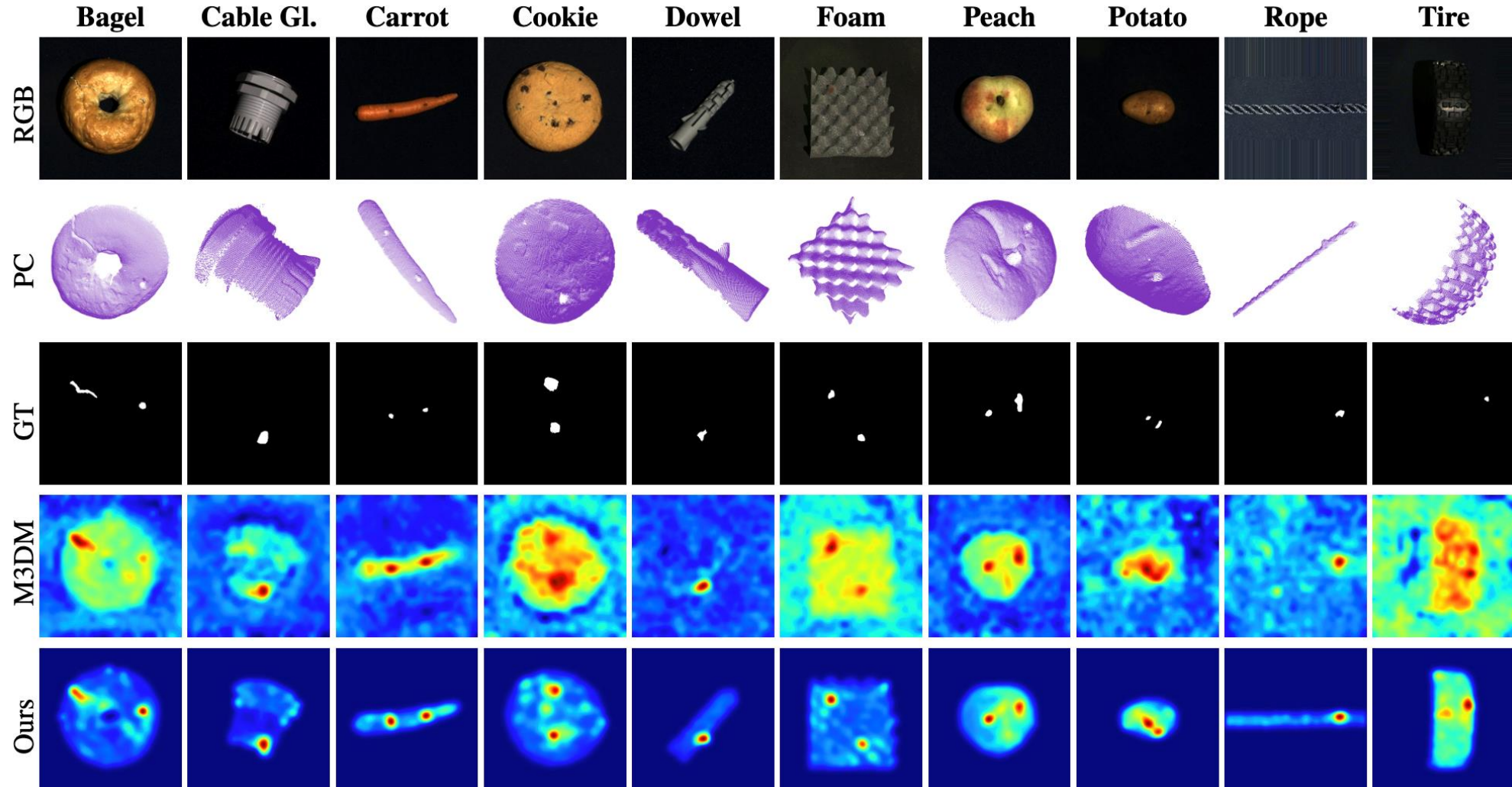


	Method	Bagel	Cable Gland	Carrot	Cookie	Dowel	Foam	Peach	Potato	Rope	Tire	Mean
I-AUROC	DepthGAN [5]	0.538	0.372	0.580	0.603	0.430	0.534	0.642	0.601	0.443	0.577	0.532
	DepthAE [5]	0.648	0.502	0.650	0.488	0.805	0.522	0.712	0.529	0.540	0.552	0.595
	DepthVM [5]	0.513	0.551	0.477	0.581	0.617	0.716	0.450	0.421	0.598	0.623	0.555
	VoxelGAN [5]	0.680	0.324	0.565	0.399	0.497	0.482	0.566	0.579	0.601	0.482	0.517
	VoxelAE [5]	0.510	0.540	0.384	0.693	0.446	0.632	0.550	0.494	0.721	0.413	0.538
	VoxelVM [5]	0.553	0.772	0.484	0.701	0.751	0.578	0.480	0.466	0.689	0.611	0.609
	BTF [17]	0.918	0.748	0.967	0.883	0.932	0.582	0.896	0.912	0.921	0.886	0.865
	AST [34]	0.983	0.873	0.976	0.971	0.932	0.885	0.974	0.981	1.000	0.797	0.937
	M3DM [40]	0.994	0.909	<u>0.972</u>	0.976	0.960	0.942	<u>0.973</u>	0.899	0.972	0.850	0.945
	Ours	0.994	<u>0.888</u>	0.984	0.993	<u>0.980</u>	0.888	0.941	0.943	<u>0.980</u>	0.953	<u>0.954</u>
	Ours-M	<u>0.988</u>	0.875	0.984	<u>0.992</u>	0.997	<u>0.924</u>	0.964	<u>0.949</u>	0.979	<u>0.950</u>	0.960
AUPRO@30%	DepthGAN [5]	0.421	0.422	0.778	0.696	0.494	0.252	0.285	0.362	0.402	0.631	0.474
	DepthAE [5]	0.432	0.158	0.808	0.491	0.841	0.406	0.262	0.216	0.716	0.478	0.481
	DepthVM [5]	0.388	0.321	0.194	0.570	0.408	0.282	0.244	0.349	0.268	0.331	0.335
	VoxelGAN [5]	0.664	0.620	0.766	0.740	0.783	0.332	0.582	0.790	0.633	0.483	0.639
	VoxelAE [5]	0.467	0.750	0.808	0.550	0.765	0.473	0.721	0.918	0.019	0.170	0.564
	VoxelVM [5]	0.510	0.331	0.413	0.715	0.680	0.279	0.300	0.507	0.611	0.366	0.471
	BTF [17]	0.976	0.969	0.979	0.973	0.933	0.888	0.975	0.981	0.950	0.971	0.959
	AST [34]	0.970	0.947	<u>0.981</u>	0.939	0.913	0.906	0.979	<u>0.982</u>	0.889	0.940	0.944
	M3DM [40]	0.970	<u>0.971</u>	0.979	<u>0.950</u>	0.941	0.932	0.977	0.971	0.971	0.975	0.964
	Ours	<u>0.979</u>	0.972	0.982	0.945	<u>0.950</u>	0.968	<u>0.980</u>	<u>0.982</u>	<u>0.975</u>	<u>0.981</u>	<u>0.971</u>
	Ours-M	0.980	0.966	0.982	0.947	0.959	<u>0.967</u>	0.982	0.983	0.976	0.982	0.972

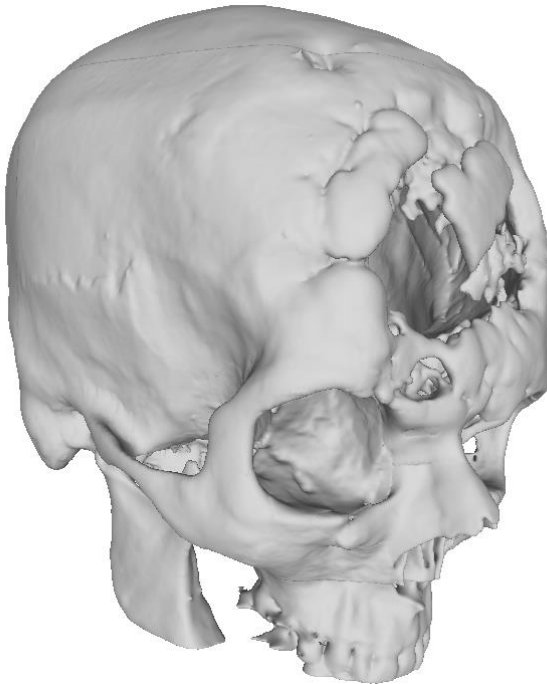
Quantitative results on MVTec 3D-AD



Qualitative results on MVTec 3D-AD



HAMLET - Head and face Automated reconstruction via Machine Learning Techniques



DIBINEM

DISI



DIMEC

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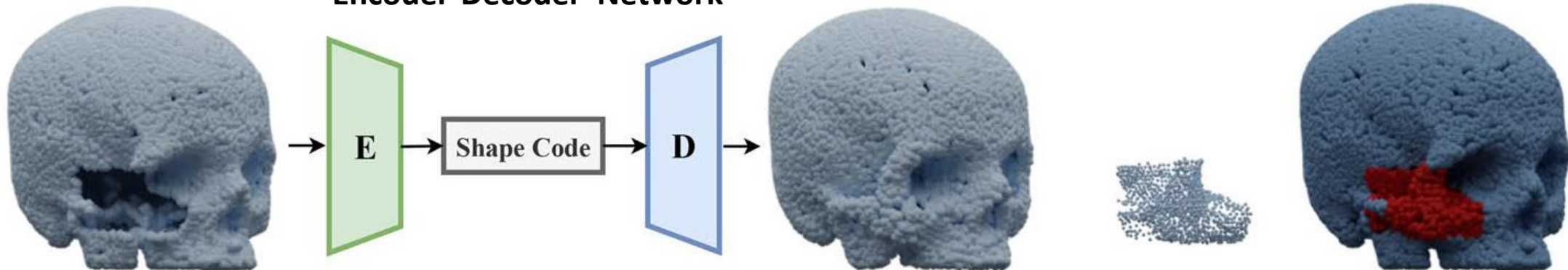
IRCCS Sant'Orsola

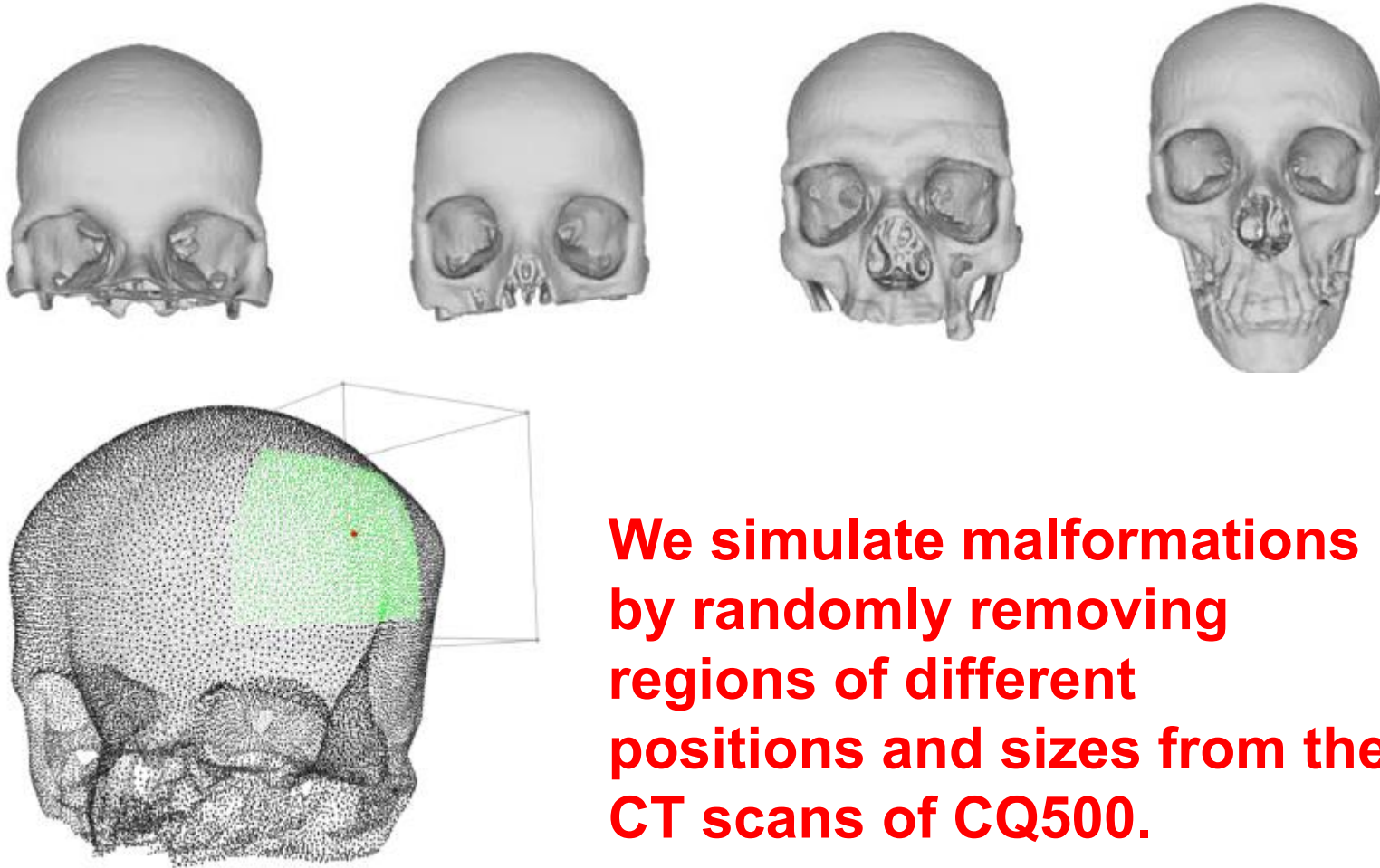


How to train the neural network ?



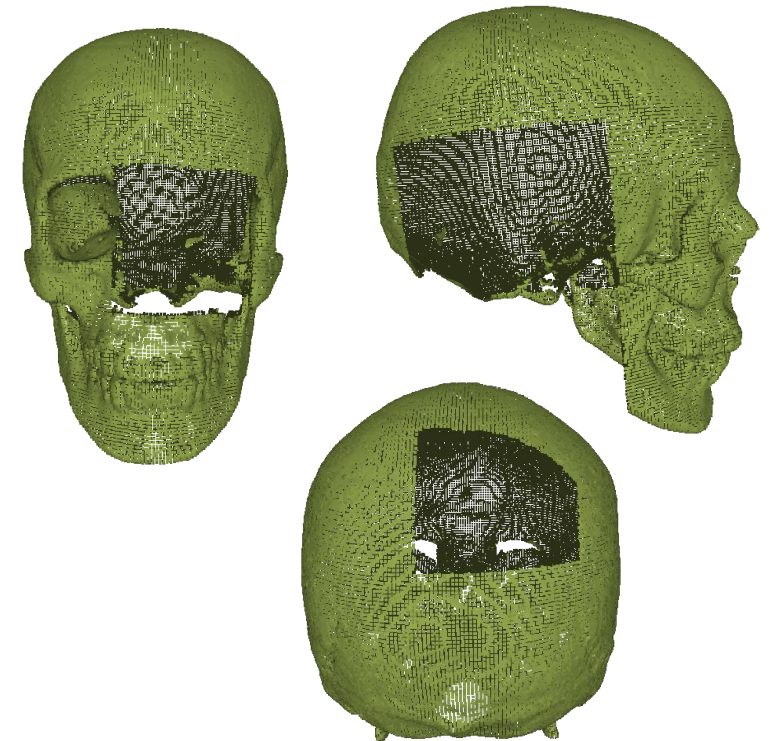
Encoder-Decoder Network



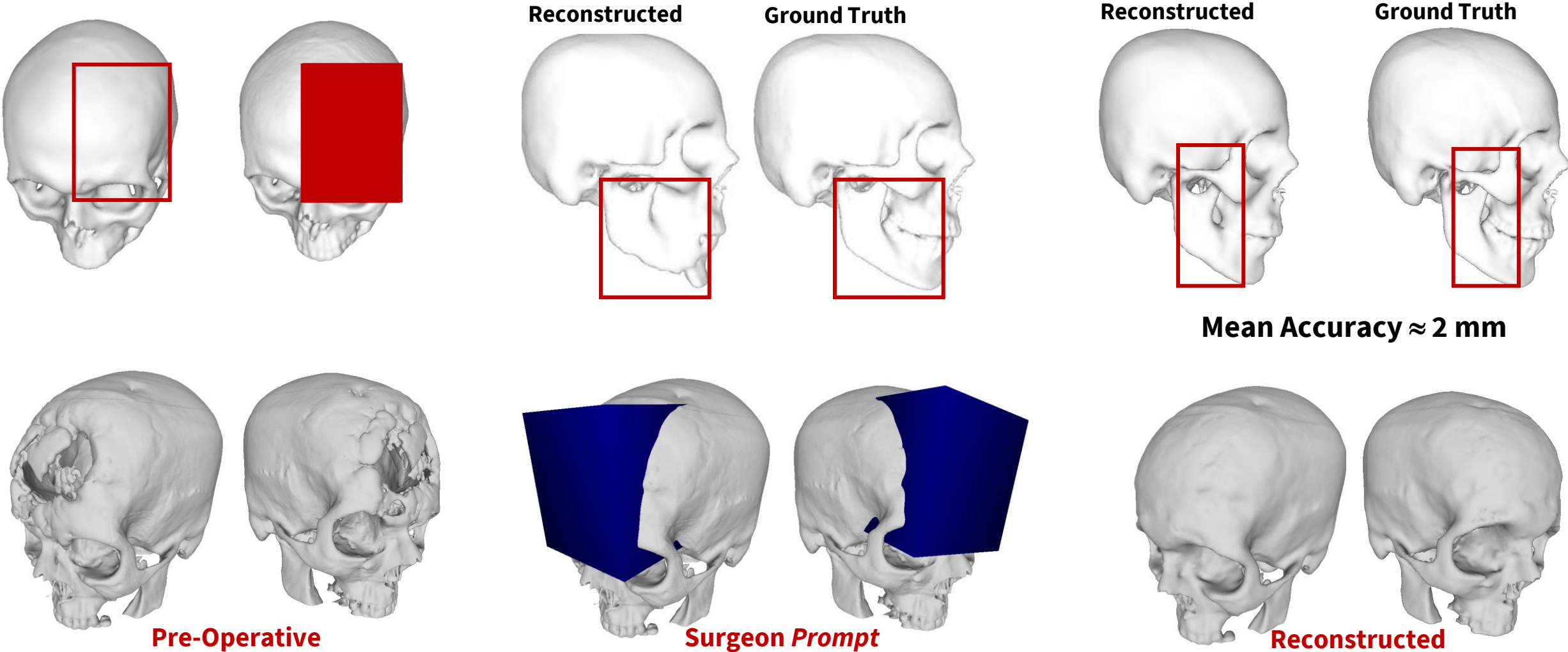


We simulate malformations by randomly removing regions of different positions and sizes from the CT scans of CQ500.

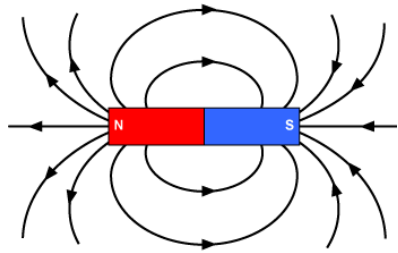
CQ500: 385 CT scans of eumorphic subjects.



Experimental Results



Fields And Neural Fields



Magnetic Field

$$\mathbb{R}^2 \rightarrow \mathbb{R}^2$$



RGB Intensity Field

$$\mathbb{R}^2 \rightarrow \mathbb{R}^3$$

Image



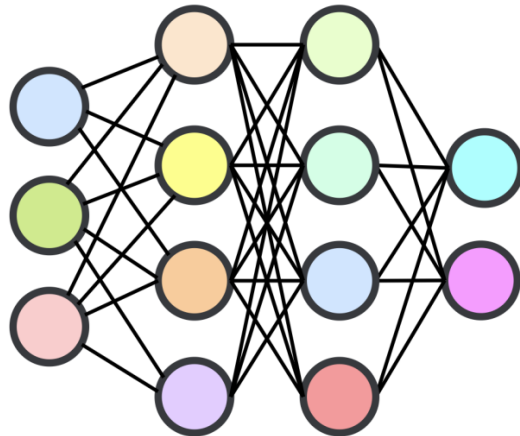
Signed Distance Field (SDF)

$$\mathbb{R}^3 \rightarrow \mathbb{R}$$

Implicit Surface

A field is a function defined for all spatial coordinates

$$x \in \mathbb{R}^n$$



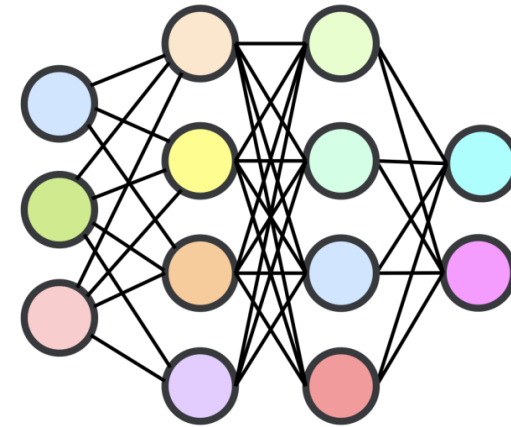
$$f(x) \in \mathbb{R}^m$$

A neural field is a field parameterized by a neural network

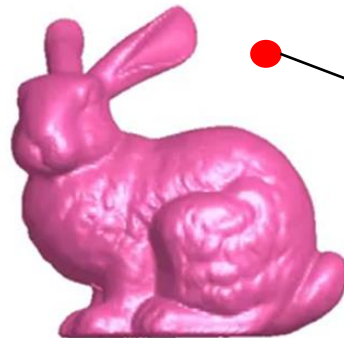
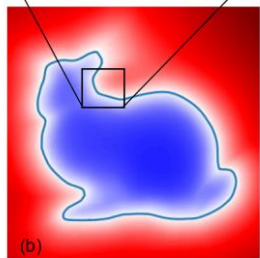
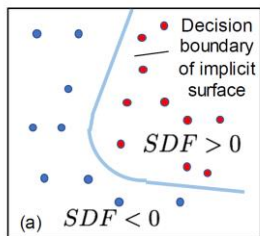
Neural Fields for Images and 3D Objects



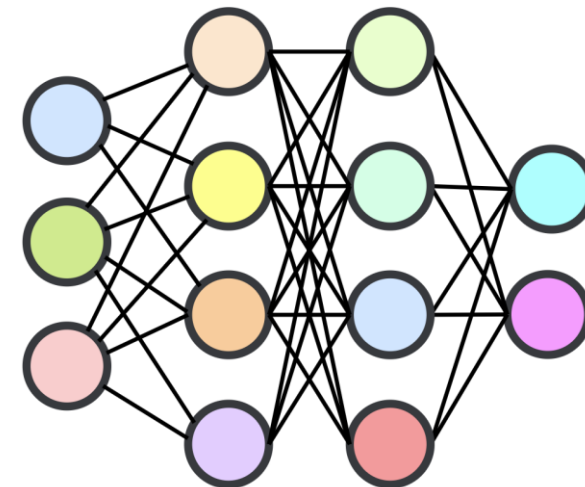
(x, y)



(R, G, B)

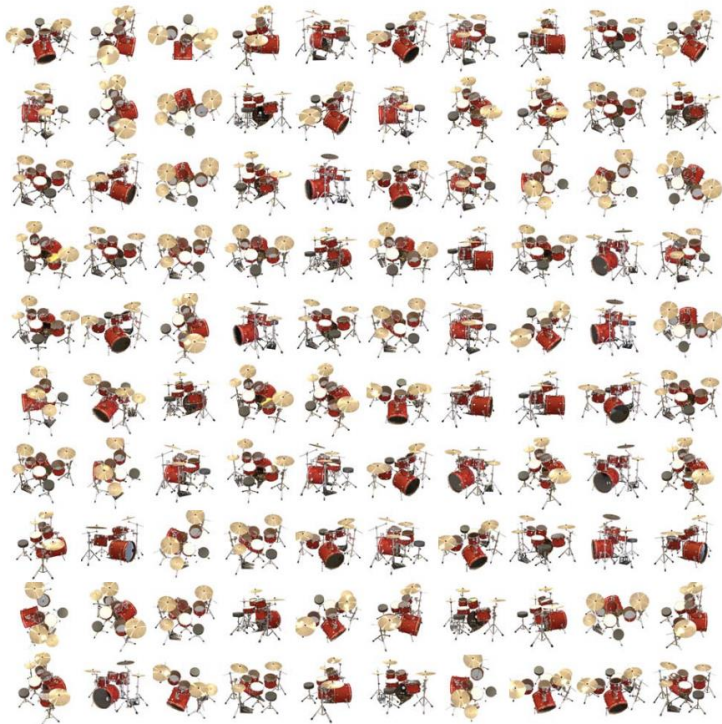


(x, y, z)

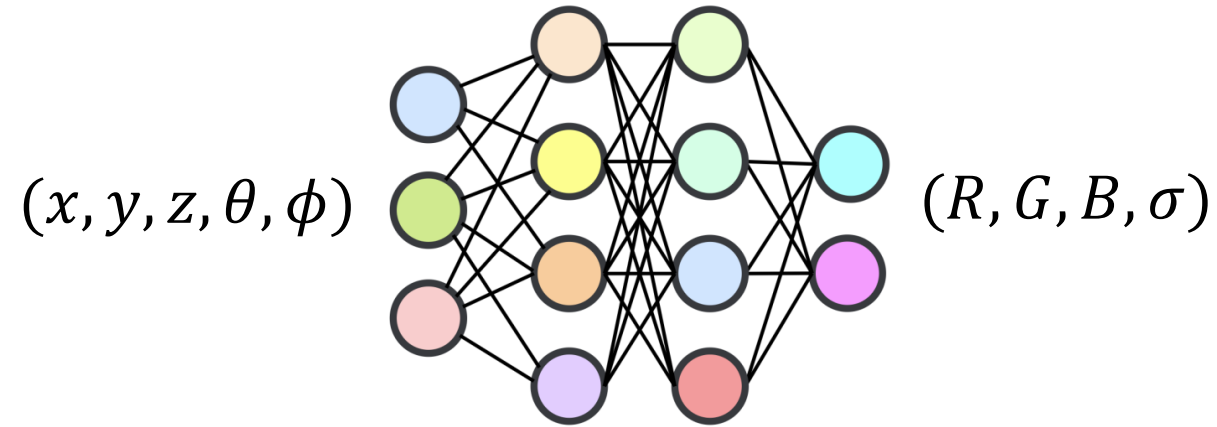


$sdf(x, y, z)$

Neural Radiance Fields (NeRFs) [1 2]



Given N (e.g. 100+) *posed* images



Train a neural network (NeRF) to predict the directional radiance and density of points in space



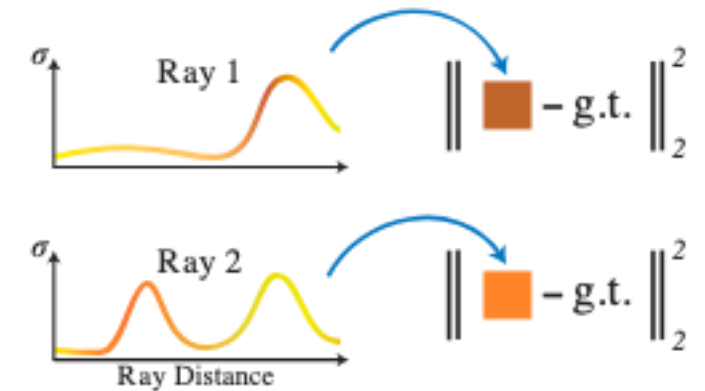
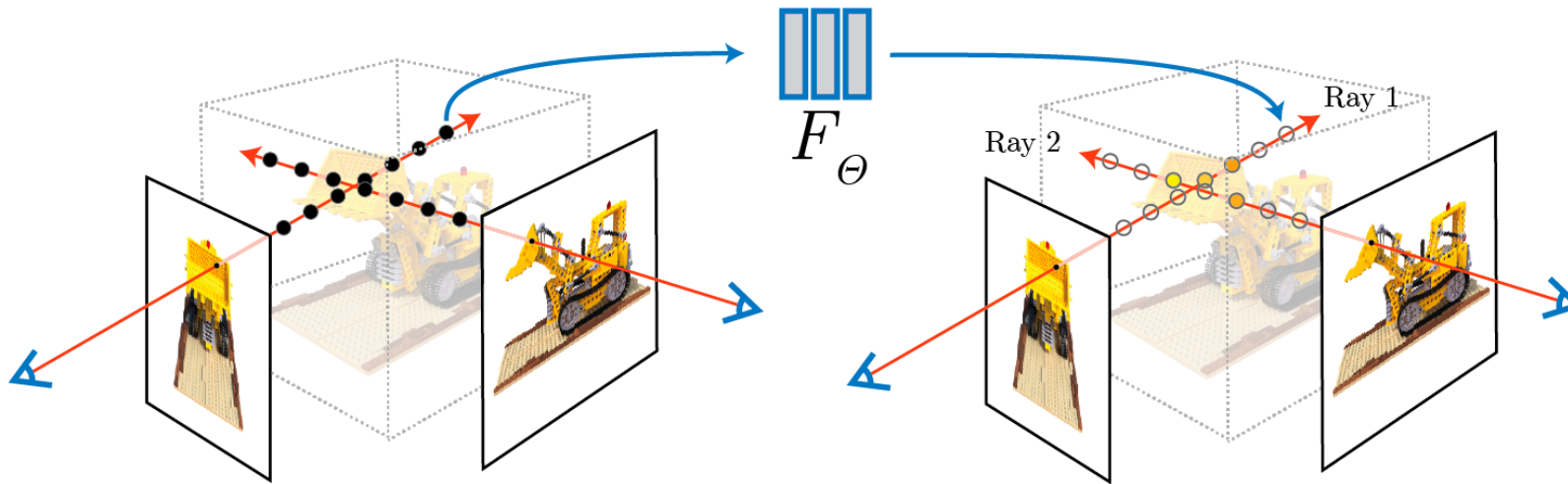
So as to synthesize novel images from any viewing direction (volumetric rendering)

How it works ? Volumetric Rendering

$$(x, y, z, \theta, \phi) \rightarrow \begin{array}{|c|c|c|} \hline \text{MLP} \\ \hline \end{array} \rightarrow (RGB\sigma)$$

F_{Θ}

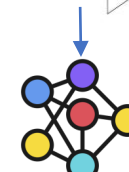
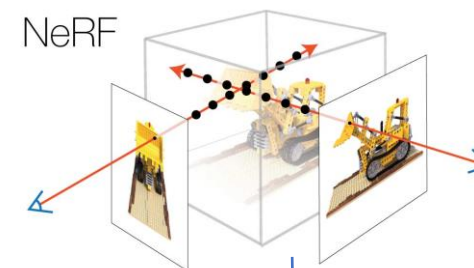
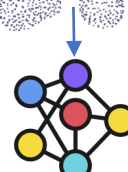
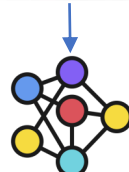
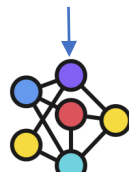
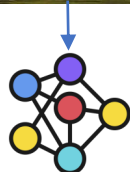
A **NeRF** is just an **MLP** that is queried with the 3D coordinates of a point in space and the viewing direction of the camera to predict a **color** and a **density** value.



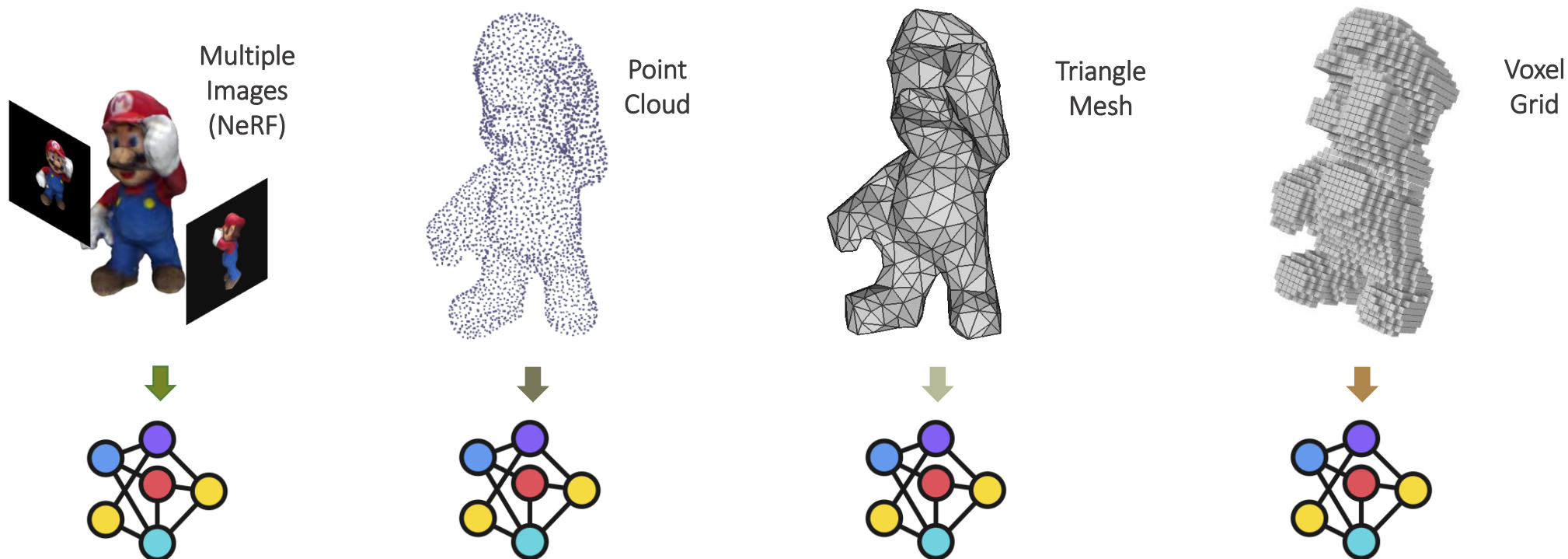
To render an image, we sample 3D points along **rays** passing through the camera center and pixels. Once color and density are predicted, they are **accumulated along the rays** to obtain pixel colors. At training time, the MLP is optimized by a **photometric loss**.

Key Advantages

- A Neural Field is a **highly compressed** and **continuous** representation which disentangles memory cost and spatial resolution. Indeed, with a fixed and small number of parameters **one can output a 3D surface or image at any arbitrarily fine resolution.**
- This representation is applicable to many signals of interest.



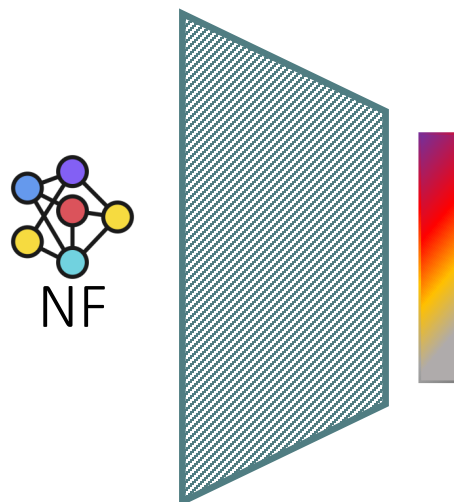
NFs are a novel representation for 3D Objects....



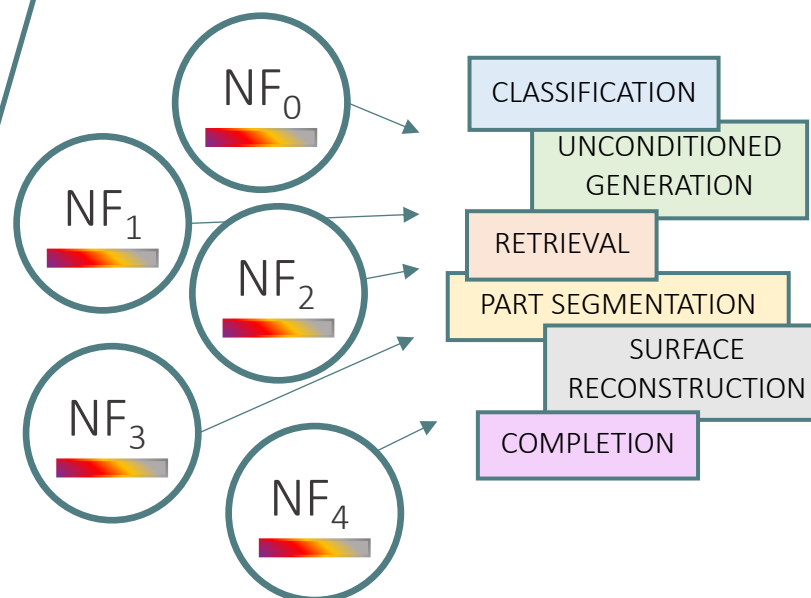
.... can we design NNs capable of processing NFs ?

nf2vec [13, 14]

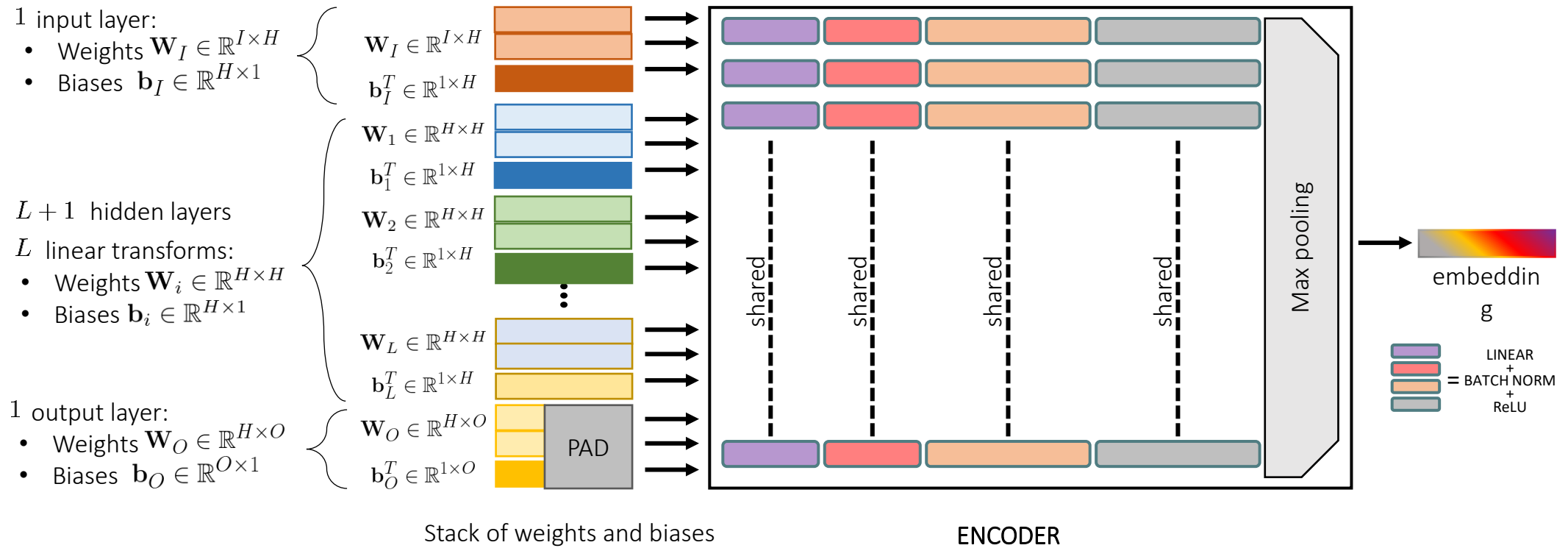
nf2vec:
embeds NFs' weights
into compact latent vectors



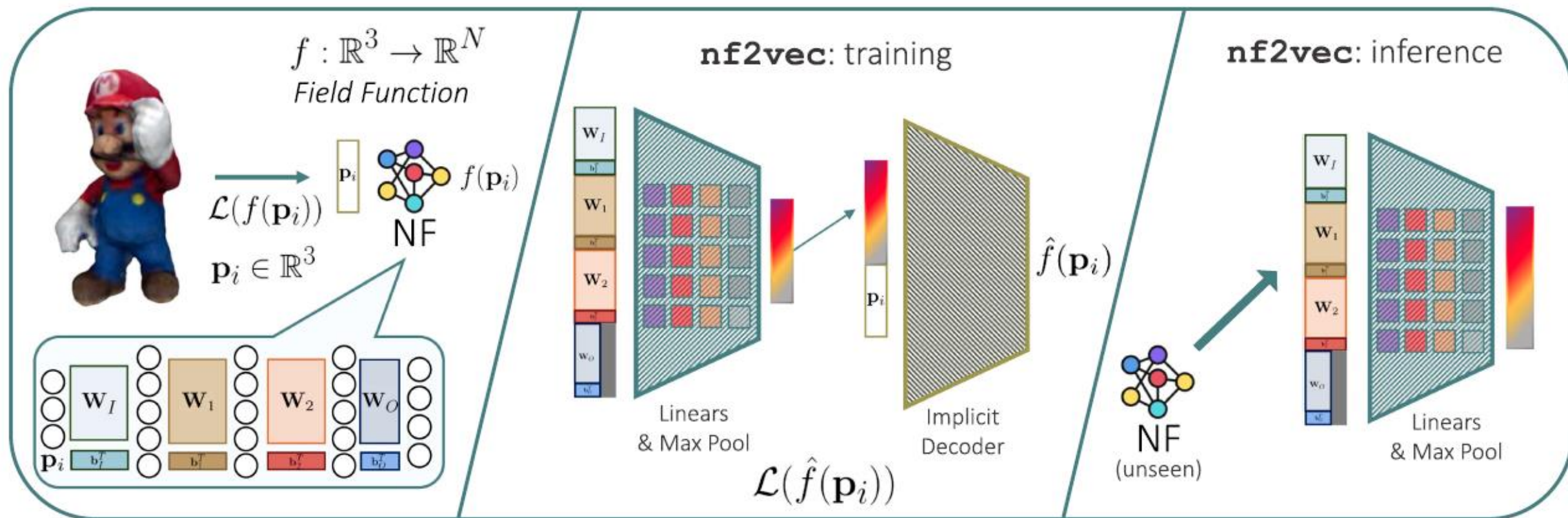
nf2vec latent vectors
can be input to standard NNs
to tackle downstream tasks



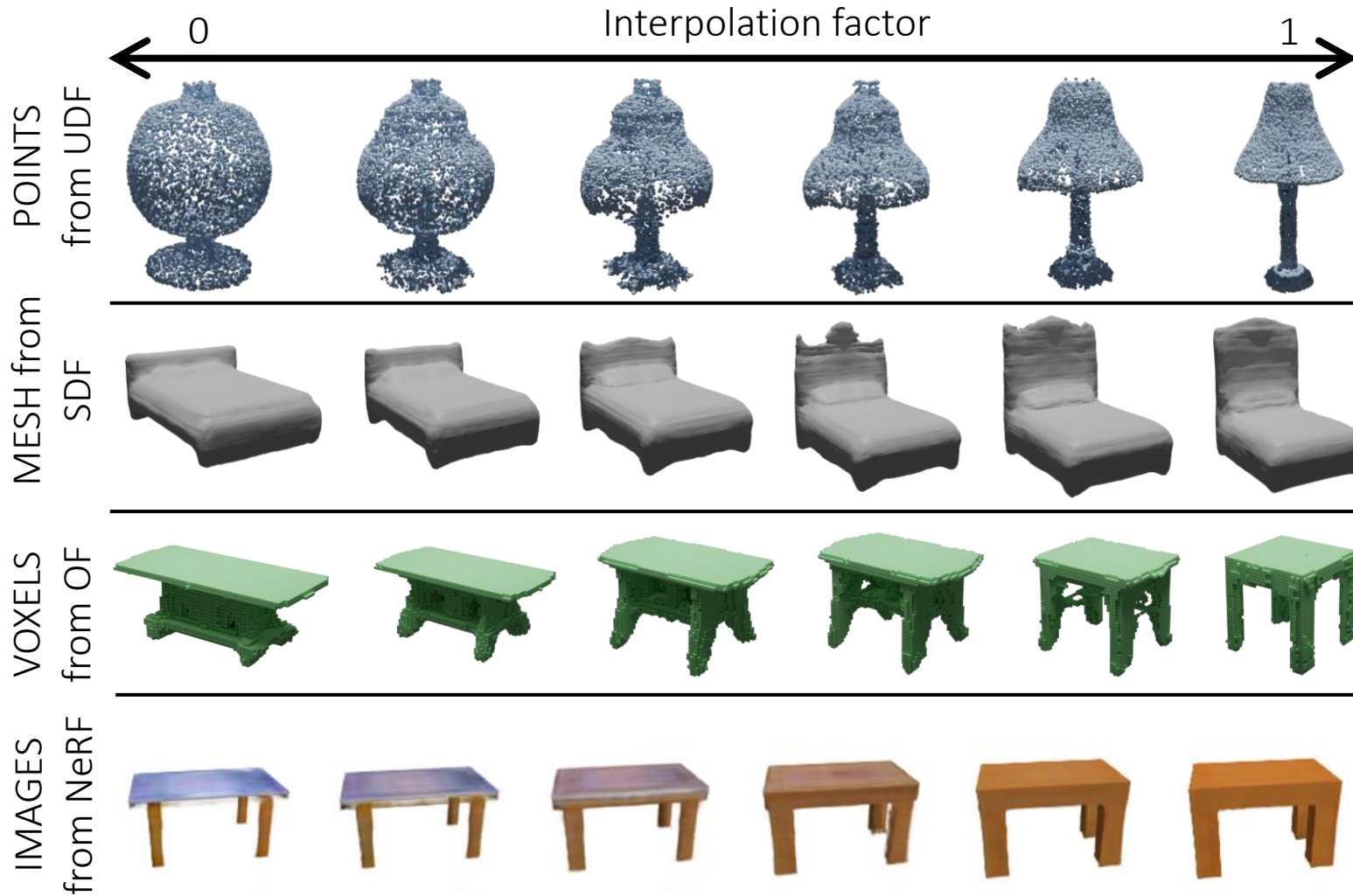
nf2vec encoder



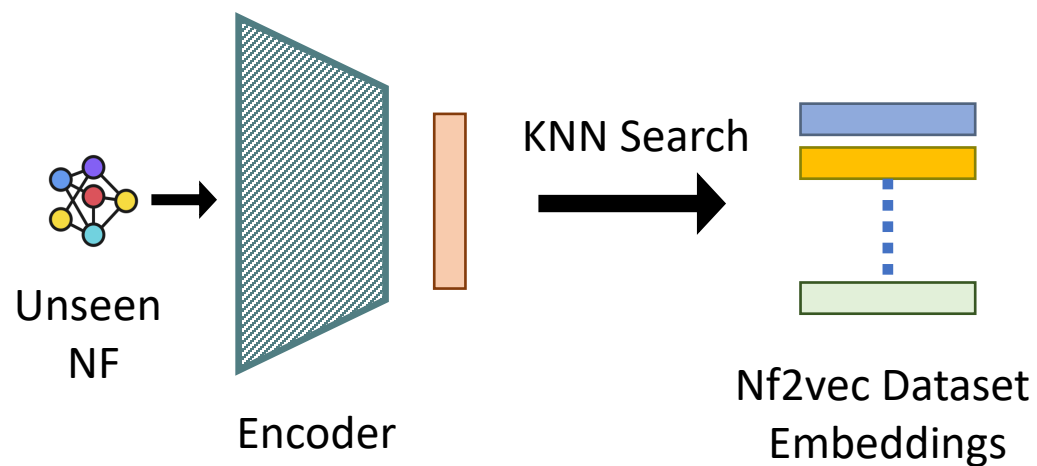
Training and Inference



Is it a good latent space ?



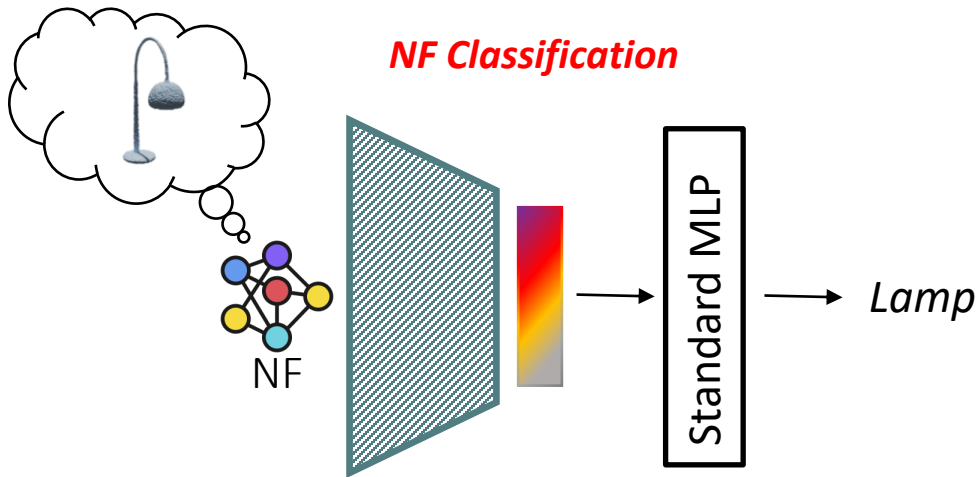
Is it a good latent space ?



Retrieval with NeRFs

Neural Processing of Neural Fields

NF Classification



Point Cloud Segmentation

NeRF Classification

Method	#Views	Accuracy (%)
nf2vec	-	87.28
ResNet50 [58] Single-view	1	86.88
ResNet50 [58] Multi-view	9	93.28

Method	Time (ms)			Total
	Encoding	Rendering	Classification	
nf2vec	0.75	-	0.28	1.03
Resnet50 [58] Single-View	-	4.63	6.21	10.84
Resnet50 [58] Multi-View	-	41.67	55.89	97.56

Generation (Point Clouds, Meshes, NeRFs) by Latent-GAN

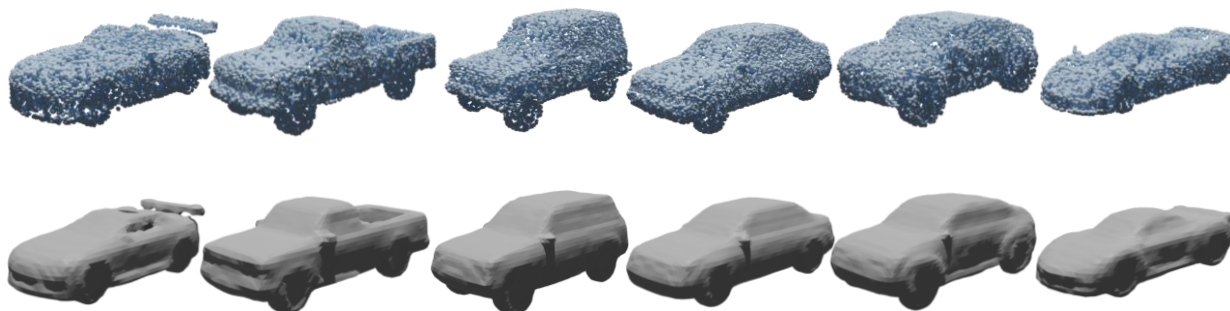


Learning Mappings between Latent Spaces

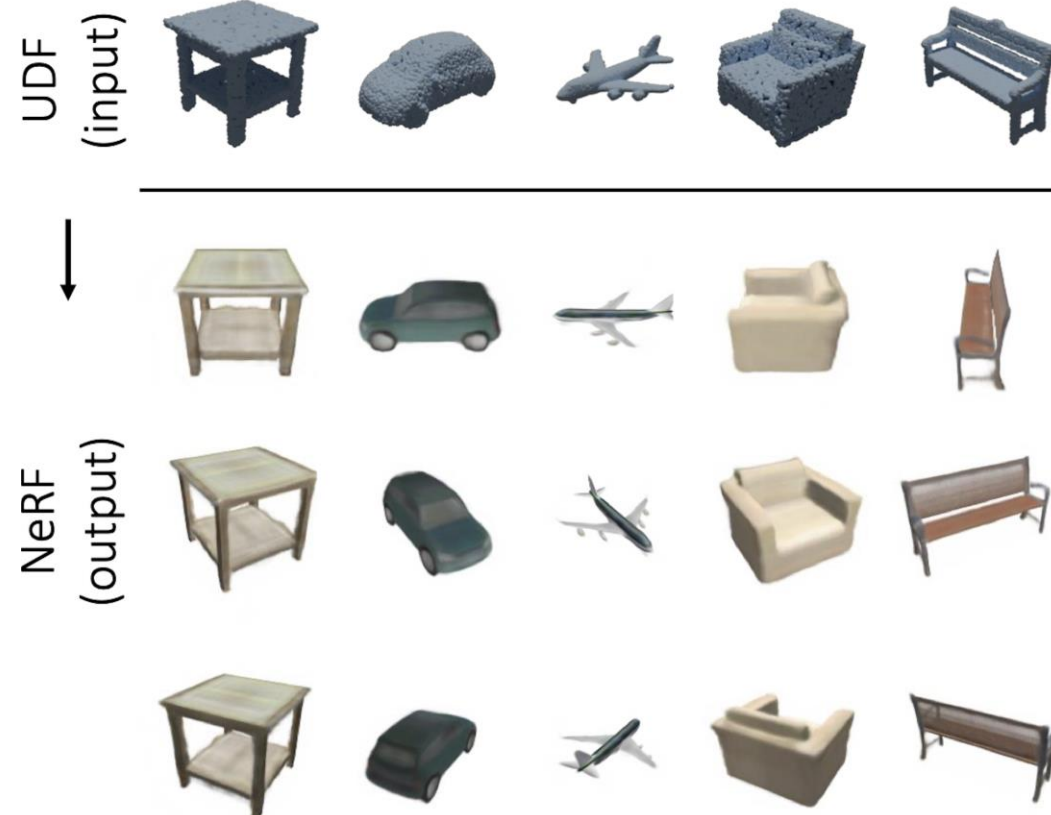
Shape Completion



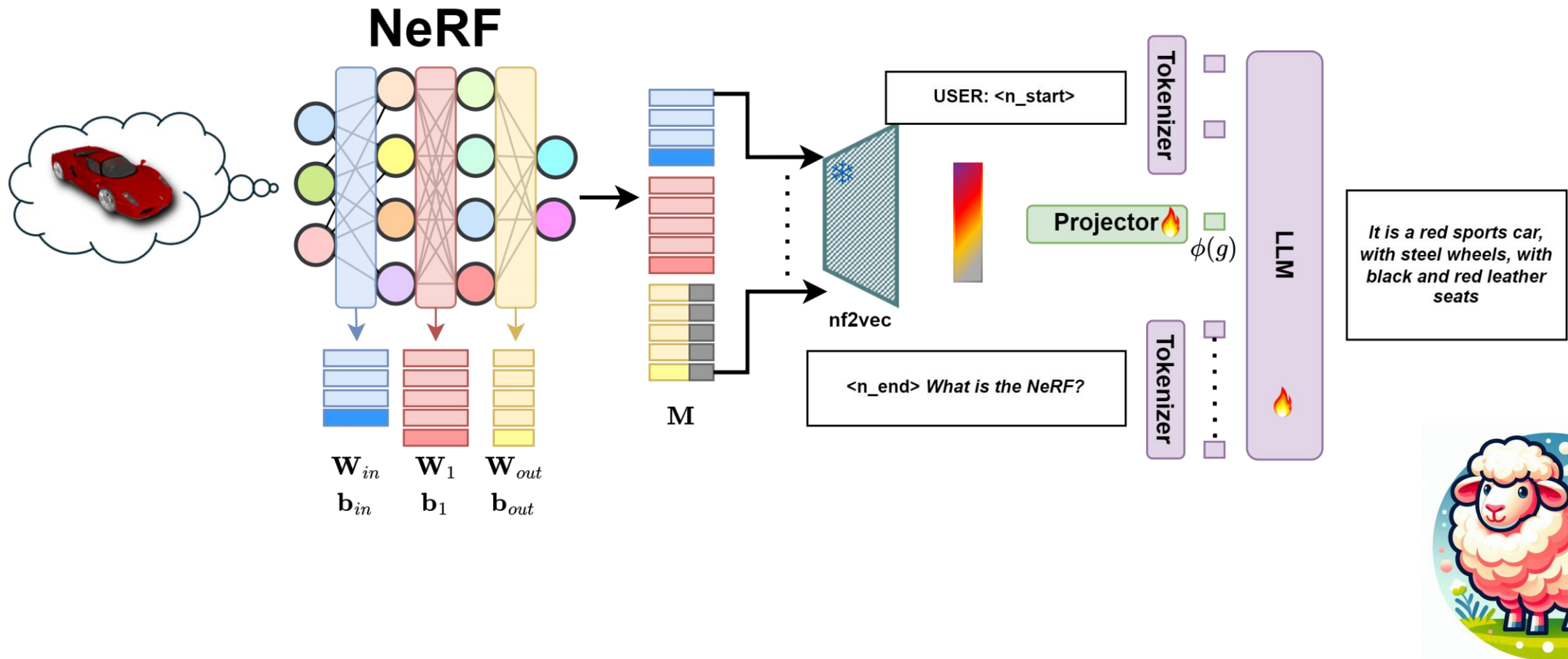
Point Cloud to Mesh



Point Cloud to NeRF

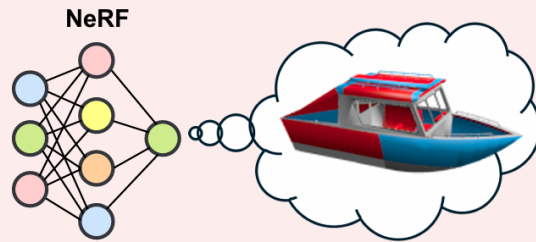


LLaNA: Large Language and NeRF Assistant [15]



LLaNA: Captioning and Question Answering

NeRF captioning



Describe the object.

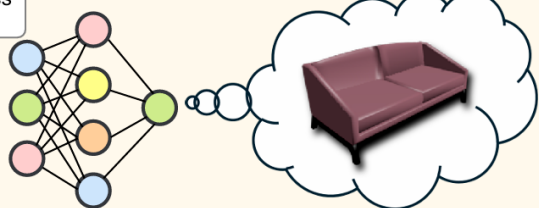


A stylized, cartoon-like depiction of a blue and red boat, characterized by a pointed front, a straight back, and a single cabin. The boat features a simplistic design with no visible textures or details that suggest material or scale, giving it a toy-like or model boat appearance. The mast is likely used for sailing or anchoring, and the overall design is reminiscent of a child's toy.

Zero-shot NeRF Classification

What is the class of the NeRF?

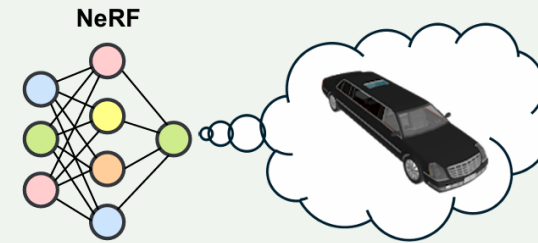
NeRF



This is a sofa.



NeRF Question Answering



What is the color scheme of the interior design of the car?



The color scheme of the interior design is dark, possibly black or grey, with a contrasting lighter color on the grille and headlights.

How can this car be used?



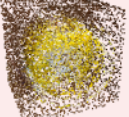
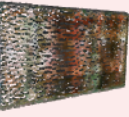
This luxury sedan can be used for transportation, providing a comfortable and stylish ride for passengers.

What is the design of the roof of the car?



The roof of the luxury sedan features a sunroof, adding to its sleek and sophisticated design.

LLaNA vs. MLLMs that can ingest renders from NeRfs

	LLaNA	PointLLM	LLaVA
Brief Description	NeRF  What object is this NeRF rendering? The object is a modern, high-end speaker with a sleek and sophisticated design. It has a rectangular shape with a brown exterior and a yellow and white circular front panel.	 What object is this point cloud rendering? The 3D model showcases a crate-like container, with a yellow, spherical object placed inside it.	 What object is this image rendering? The image shows a 3D model of a box with a circular emblem on it. The emblem is yellow with a white border and features the letters "JJA" in the center.
Detailed Description	NeRF  Elaborate on the details of this NeRF, please. The object is a flat-screen TV with a sleek and modern design, featuring a black frame and a thin bezel. The TV has a stand at the bottom center, made of a combination of white and black materials, with a curved design that adds a touch of elegance. The TV screen is rectangular in shape, with a black border surrounding it. A small, centered logo is located on the bottom right corner of the TV, adding a touch of sophistication to its overall design.	 Elaborate on the details of this point cloud, please. This 3D object model represents a cartoonish image featuring a bright, blue background. Overlaying this is an image of a desert landscape. The juxtaposition of the vibrant background with the depiction of a desert creates an interesting visual dynamic. The desert scene is not detailed, rather it has a stylized, simplistic aesthetic, mirroring the playful, cartoon-like vibe of the background. This object might be used in graphic design, animation or other related fields to evoke a sense of whimsy, fantasy, or to portray various concepts.	 Elaborate on the details of this image, please. It seems to be a collection of images or a collage of faces. The images are arranged in a grid-like pattern, with each image occupying its own space. The images are likely portraits, given the presence of faces. The images are of varying sizes, with some being larger and more detailed than others. The overall style of the image suggests it could be a digital artwork or a representation of a digital interface.
Single-Round Q&A	NeRF  What is the design of the wheels? The wheels are fitted with a five-spoke alloy design.	 What is the design of the wheels? The wheels have a black mesh-style design.	 What is the design of the wheels? The image you've provided is too small and blurry to discern any specific details about the wheels or the design of the car.



Thanks a lot
for your
attention !

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